

- Introduction
- For Your Safety
- Description
- Installation
- Startup
- Diagnosis
- Glossary
- Legal Provisions

MANUAL ENGLISH

for the Devices of the Cube67+ BN-EtherCAT Series
Art. No. 56527

Publisher's Note

Product Manual for Cube67+ BN-EtherCAT (Article Number: 56527)

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1 Introduction / About This Manual

1.1 Service and Support

Sales

You can rely on the support of our sales staff in internal sales and field sales at all times.

CONNECTIVITY System Advisor

Our system advisors are your competent contacts when it comes to developing CONNECTIVITY solutions. They cooperate with you to find the best solutions for your electrical installations. They support and advise you during the planning, conception and project engineering phases for your application – with professional competence. They will also visit you to advise you at your company.

CONNECTIVITY advisors cooperate with you to find ways boost the competitiveness of your machines and systems in the long term.

Customer Service Center (CSC)

Our Customer Service Center staff help you with all queries relating to installation and commissioning. For example, they support you with problems which deal with the compatibility of products from different hardware and software manufacturers.

They are backed by numerous support tools, as well as testing equipment for fieldbus systems and EMC impacts.

Call us on +49 (0) 7191 47-2050, or send an email to

csc@murrelektronik.de

1.2 About This Manual

Information in this document

- ➔ Read this section carefully before you start working with the documentation or the device.
- ➔ Read the documentation carefully before starting up the device.
- ➔ Store the manual in a place that is accessible to all users at all times for the entire service life of the device.

You will need general knowledge about automation engineering in order to understand this manual. In addition, planning and using automation systems requires technical knowledge which is not contained in this manual.

Function of this manual

This manual describes how to use the Cube67+ BN-EC, Art. No. 56527.

This manual instructs the technical staff of the machine manufacturer or machine operator on how to install, configure, start up, wire up, and operate the bus node. It is ***not*** a description of how to operate the machine in which the bus node is integrated. For information about this, please refer to the operator manual belonging to the machine.

Applicable Documents

Document	Art. No.	Location
Installation Instructions	56527	Murrelektronik GmbH online shop http://www.murrelektronik.com/
Cube 67/Cube67+ Manual Technical Data	56971	
Cube67+ System Manual	56974	
Cube20 System Manual	56030	
Cube20 System Manual Expansions	56035	
Bus Manual		 www.ethercat.org
EtherCAT Specifications	ETG.1020	

1.3 Icons Used In This Manual



DANGER

Immediate danger

- Failure to observe this warning will result immediately in death or severe personal injury.



WARNING

Possible danger

- Failure to observe this warning may result in death or severe personal injury.



CAUTION

Low risk hazard

- Failure to observe this instruction will lead to personal injury of light to medium severity.

CAUTION

Risk of material damage

- Failure to observe this warning will lead to material damage.



NOTE

Further technical information and notes provided by Murrelektronik GmbH.



RECOMMENDATION / INFORMATION

References to additional literature, tips, and recommendations provided by Murrelektronik GmbH.

Action

- ➔ An arrow indicates action to be taken.
- ➔ Read and follow the action to be taken.

Hyperlink, jump command

[Text in blue indicates a jump command.](#)

1.4 Trademarks

EtherCAT®
TwinCAT®

Registered trademarks of Beckhoff Automation GmbH

2 For Your Safety

For your safety and the safety of machine operators

- ➔ Read this section carefully.
- ➔ Only then should you work with the bus module and the Cube 67 system.

2.1 General Safety Instructions

About this product

The product described in this manual was developed, manufactured, tested, and documented in compliance with the relevant safety standards. Provided you comply with the handling instructions and the safety instructions described here, this product will not normally cause any hazards to persons and assets. The EtherCAT bus model meets the specifications of European EMC Directive (2004/108/EC).

Qualified staff

The EtherCAT bus module may only be fitted, put in service, and operated by qualified staff.

They are qualified if they meet the following requirements:

- They have the relevant technical training.
and
- They were trained by the machine operator to operate the machine and received instruction in the related safety instructions.
and
- They have access to the Installation Instructions and the manual.
and
- They are familiar with the safety standards which are common practice in automation engineering.

Use of the module

- ➔ Comply with all safety and accident prevention regulations when conducting project engineering, installation, startup, operation, and testing of all devices.
- ➔ Verify material resistance if aggressive media are used.

Only Murrelektronik technical personnel are allowed to conduct work on the module hardware and software, if this is not described in this manual.

2.2 Intended Use

Intended Use

The EtherCAT bus module is a module in the decentralized Cube67 and Cube67+ families. It can be used in tough industrial environments up to IP67 Protection Class.

The intended use of the module and the IP67 Protection Class is only guaranteed if all connections are hooked up using connectors or screw plugs.

The intended use also includes the electrical installation in compliance with EMC regulations. The EtherCAT bus module is designed for use in an industrial environment. Radio interference may occur when the module is installed for domestic use.

- If you install the EtherCAT bus module in a residential home, please comply with the related standards.

Foreseeable misuse

- Do not alter the design, engineering, or electrical features of the EtherCAT bus module.
- Do not use the EtherCAT bus module outside of the application areas described in this manual, the Technical Data, and the Installation Instructions.
- Do not use the EtherCAT bus module as a safety device. It does not comply with the related standards. The safety functions of the system are then no longer guaranteed.
- Do **not** use the OFF state of the EtherCAT bus module outputs to send safety-related signals to the machine.
- Do **not** use the EtherCAT bus module outdoors, or in continuous operation in liquids.
- Do **not** clean the EtherCAT bus module using high-pressure tools.

2.3 Warranty and Liability

Warranty and liability claims become null and void

Warranty and liability claims become null and void if

- the product is not used for its intended purpose.
- improper use results in damage, or if the user or operator fails to comply with the manual or Installation Instructions,
- or if the personnel is/was not qualified.

2.4 Ecological Disposal

Comply with country-specific waste disposal regulations

- Dispose of scrap module in compliance with the valid country-specific regulations for waste disposal (e.g. the European Waste Code 16 02 14).

Only qualified persons may sort scrap materials.

- The bus module is encapsulated. Take care when dismantling the module since you could injure yourself.
- Sort the separated components into the correct recycling line.

3 Description

3.1 Bus Technology

3.2 Topology

Line topology

The bus node and its expansion modules can be integrated in an EtherCAT network in line topology.

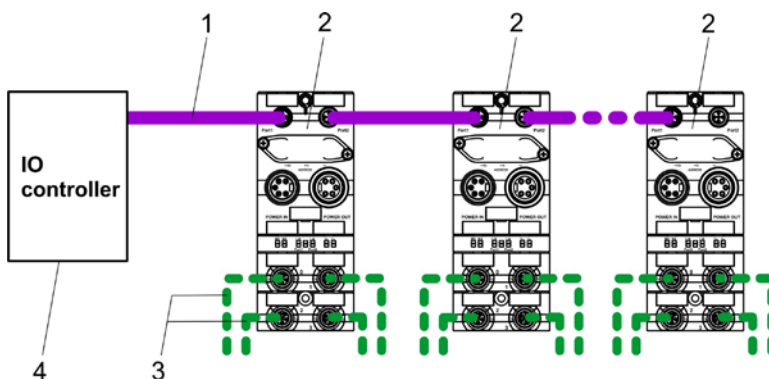


Fig. 1 EtherCAT system design

Legend

- 1 EtherCAT network
- 2 Cube67+ BN-EC bus module
- 3 Link to expansion modules
- 4 EtherCAT master

3.2.1 EtherCAT

Functional principle

From the viewpoint of the Ethernet, an EtherCAT bus is a single Ethernet user. This Ethernet user receives and sends telegrams. The Ethernet user contains a large number of EtherCAT slaves, but no Ethernet controller.

The EtherCAT slaves process incoming telegrams. They extract the useful data addressed to them and forward the telegram to the next EtherCAT slave.

The last EtherCAT slave sends the fully processed telegram back, i.e. this telegram represents a reply telegram which the first EtherCAT slave returns to the controller.

An EtherCAT user exploits the feature that Ethernet transfers data separately in the go and return directions (Tx and Rx lines) and operates in full-duplex mode.

3.3 Systems

Cube67

Cube67 is a modular decentralized I/O system which combines the IP20 and IP67 Protection Classes. The modules are pluggable, robust, and fully encapsulated. Starting from the bus module, the I/O layer extends in a star shape within the application. Hybrid system cables interconnect the power supply and communication between individual system components.

The Cube67 System comprises I/O modules for digital, analog, and serial signals, temperature detection, counters, and to couple valve islands, drives or operator panels. It offers continuous, channel-specific diagnoses down to sensor and actuator level. The digital channels are freely parameterizable. A slot or a signal terminal then becomes an input or an output.

Cube67+

The features of the Cube67+ System are extended, compared with the Cube67 System.

Cube67 and Cube67+

Cube67 and Cube67+ help you to simplify your project engineering effort, reduce installation work, shorten startup, make troubleshooting easier, and minimize downtimes.

3.4 Internal System Connection

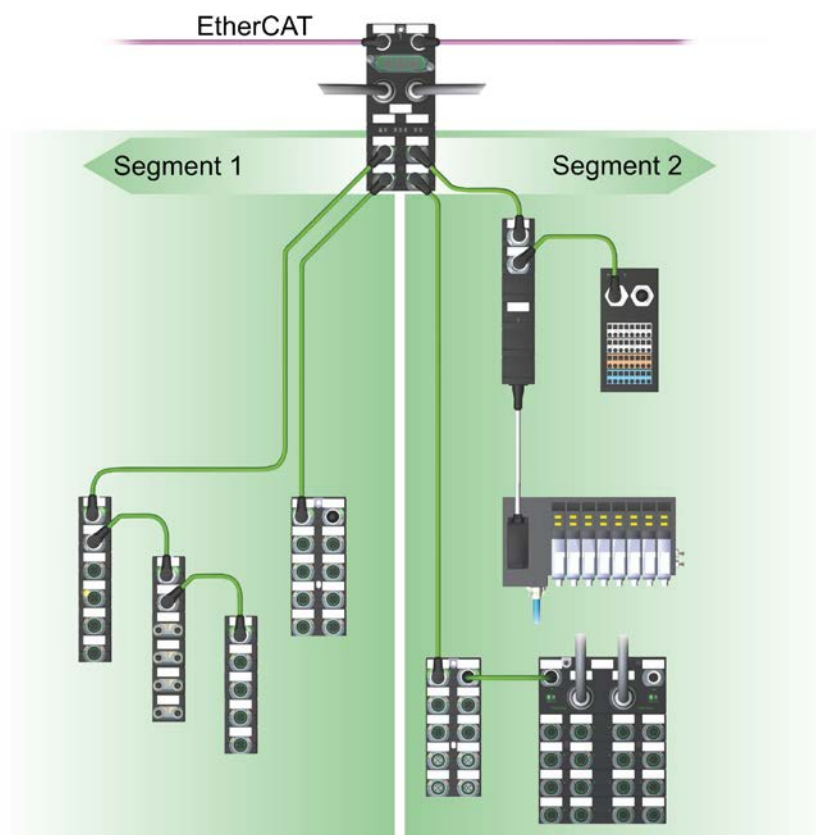


Fig. 2 *Internal system connection features*

Legend

Segments 1, 2

For each segment:

Line length: max. 30 m

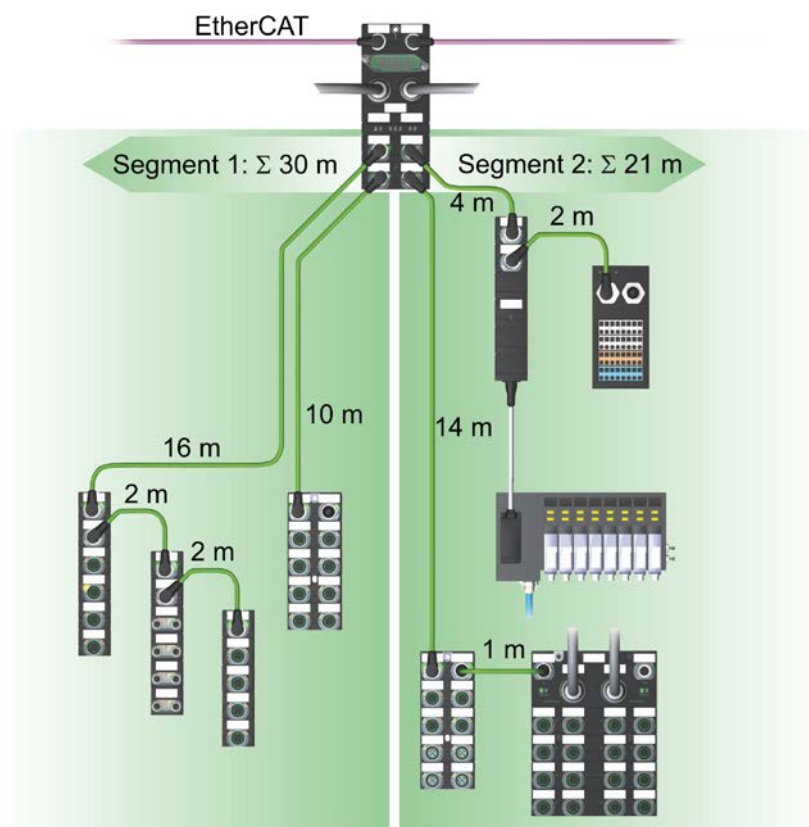
Modules max. 16

Internal system connection

The internal system connection is divided into 2 segments and, as a result, may be operated with cables of up to 30 m in length and with max. 16 modules per segment.

Segment assignment

	Segment 1 left	Segment 2 right
Sockets 0, 2	X	
Sockets 1, 3		X
Max. line length	30 m	30 m
Subdivision of line lengths	as required	as required
Max. number of modules	16	16
Module distribution	as required	as required

Tab. 1: *Segment assignment***Example: Segments**Fig. 3 *Example: Segments*

Lines

When modules are connected to socket x, this is referred to as connection to line x, where x is the socket number.

Example:

Line 0 to socket 0

Line 2 to socket 2

3.5 Bus Nodes

Cube67+ BN-EtherCAT Art. No. 56527

The EtherCAT bus module collects and distributes data to the connected Cube67 modules.



3.5.1 Features

Parameterization in the EtherCAT master

The EtherCAT bus node and the connected Cube67 modules are parameterized in the EtherCAT master.

Signal response times

The signal response time is independent of the number of I/O modules connected.

Signal response time at inputs

Signal response time for the inputs of the connected Cube67 modules **5 ms**

Delay time of input filter

Input filter delay time considered **1 ms**

Signal response time of outputs

Signal response time for the outputs of the connected Cube67 modules **4 ms**

Usable Modules

The EtherCAT **Cube67+ BN-EC** bus node permits the operation of Cube67 and Cube20 modules. The connectable modules are listed in Annex 7 [Annex 1: Supported Modules](#) and Accessories

3.5.2 Technical Data

EMC - EN 61131-2 Product Standard	EN 61000-4-2 ESD	Contact ± 4 kV, air ± 8 kV
	EN 61000-4-3 RF Field & GSM	10 V/m
	EN 61000-4-4 Burst	± 2 kV
	EN 61000-4-5 Surge	asym./symm. ± 500 V (DC input) asym. ± 1 kV (signal connections)
	EN 61000-4-6 HF asymmetrical	10 V
	EN 61000-4-8 Magnetic field 50 Hz	30 A/m
	EN 55011 Emission	QP 40 dB μ V/m (30 to 230 MHz) QP 47 dB μ V/m (230 to 1000 MHz) (Class A)
Ambient Conditions	Standard operating temperature	0°C to +55°C
	Storage temperature	-25°C to +85°C
	Housing Protection Class acc. to EN 60529	IP67
Mechanical ambient conditions	Vibration as per EN 60068 Part 2-6	5 to 70 Hz; const. amplitude 0.75 mm 70 ... 500 Hz; const. acceleration 15 g
	Shock as per EN 60068 Part 2-27	Amplitude 50 g, 11 ms duration
Connections	Power Supply Line	7/8" connector
	Bus terminal	2 x M12 socket 4-pin, D-coded
	Internal system connection	4 x 6-pin, M12 connector
Miscellaneous	Dimensions (LxWxH)	151x62x40.5 mm
	Weight	approx. 360 g
Bus data	Transmission protocol	EtherCAT
	Transmission rates	10/100 Mbit/s, IEEE 802.3, Auto-negotiation, semi or full duplex at 10 and 100 Mbit/s, automatic settings
	Electrical insulation	500 V between bus and internal logic
Power supply	Operating voltage U_S and sensor power supply 24 VIN	24 VDC (must always be connected)
	Actuator power supply 24 V	24 VDC
	Current per pin	max. 8 A
	Operating voltage range	18 to 30 VDC
	Power consumption	≤ 200 mA
	Sensor power supply	24 VDC (not switchable)
	Operating voltage range of sensor power supply	18 to 30 VDC
	Actuator power supply	24 VDC (switchable)
	Operating voltage range of actuator power supply	18 to 30 VDC

	Reverse polarity protection of module electronics	Yes
	Reverse polarity protection of sensor power supply	Yes
	Reverse polarity protection of actuator power supply	Yes
	Overvoltage protection	Yes (suppressor diode)
Internal system connection	Rated current of sensor power supply	4 A per line
	Rated current of actuator power supply	4 A per line
	Overload / short-circuit	Electronic short-circuit recognition Recognition time < 10 ms

3.5.3 Device Design

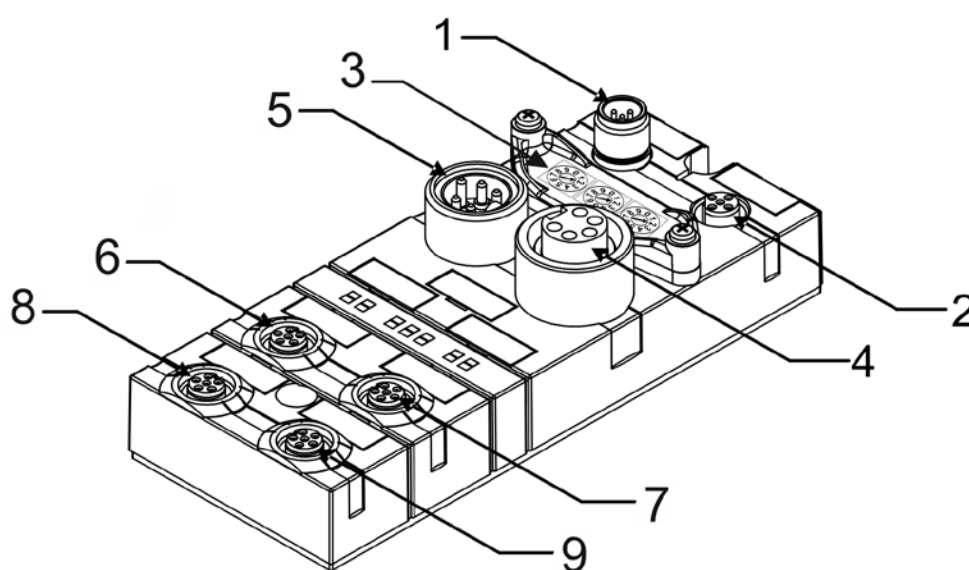


Fig. 4 Device design

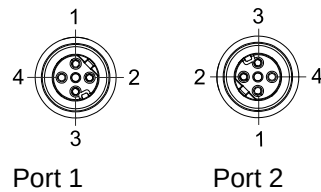
Legend

No.	Assignment	Description	Connection
1	Port 1	EtherCAT bus	M12 socket
2	Port 2	EtherCAT bus	M12 socket
3	Rotary switch	Addressing device ID	
4	Power OUT	Power supply	7/8" socket
5	Power IN	Power supply	7/8" plug
6	0	Internal system connection	M12 socket
7	1	Internal system connection	M12 socket
8	2	Internal system connection	M12 socket
9	3	Internal system connection	M12 socket

3.5.4 Connections

Port 1, Port 2

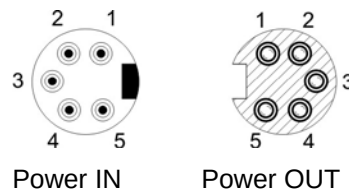
M12 sockets
EtherCAT bus



Pin 1 TD +
Pin 2 RD +
Pin 3 TD -
Pin 4 RD-

POWER IN, POWER OUT

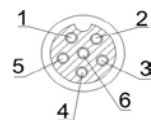
7/8" connector



Pin 1 0 V
Pin 2 0 V
Pin 3 
Pin 4 +24 V U_S / 8 A
Pin 5 +24 V U_A / 8 A

Cube67 System

M12 sockets



Pin 1 +24 V U_A / 4 A
Pin 2 +24 V U_S / 4 A
Pin 3 0 V
Pin 4 internal bus
Pin 5 internal bus
Pin 6 0 V

3.5.5 Indicators

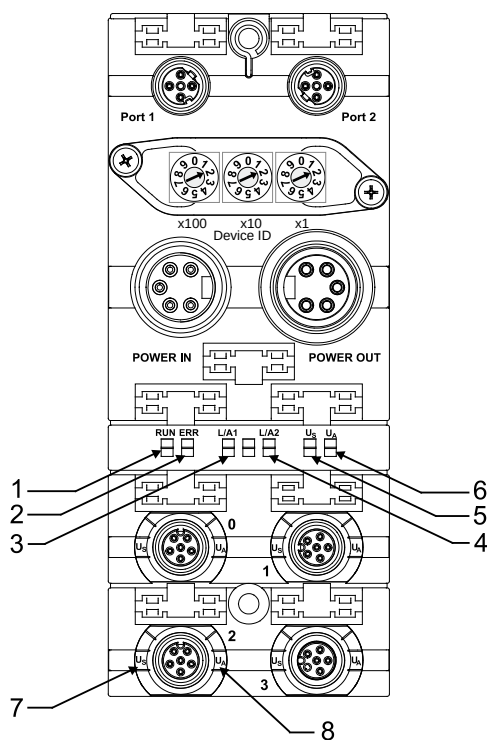


Fig. 5 LED Indicators

Legend

1	RUN	Status LED of EtherCAT state machine of Cube67+ Systems
2	ERR	Error display
3	L/A1	Link/Activity LED indicates present state of Ethernet connection at Port 1.
4	L/A2	Link/Activity LED indicates present state of Ethernet connection at Port 2.
5	U _S	LED sensor power supply
6	U _A	LED actuator power supply
7	U _S	Power supply and communication states of U _S LEDs at sockets of internal system connection
8	U _A	Power supply and communication states of U _A LEDs at sockets of internal system connection

4 Installation

4.1 Mounting

4.1.1 Dimensions

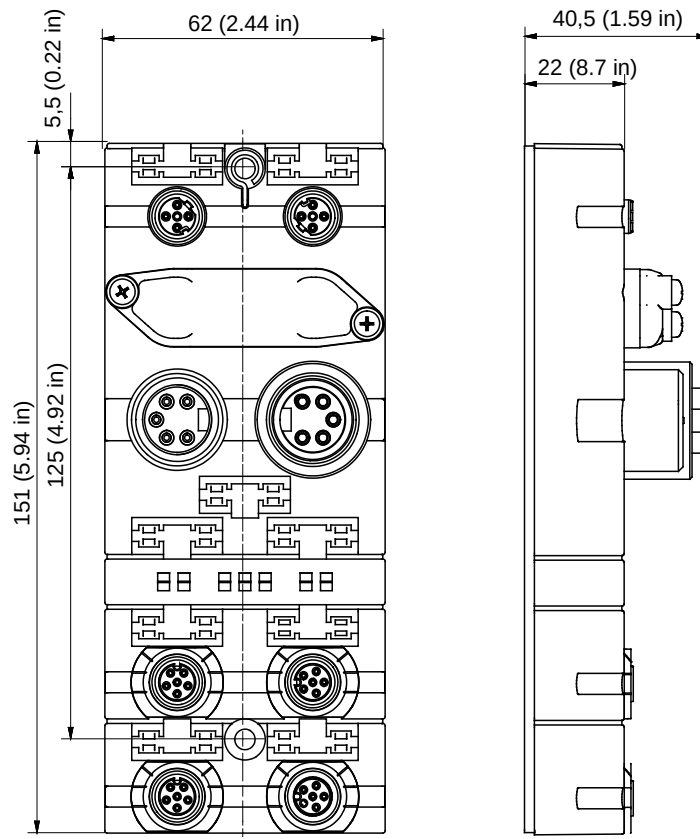


Fig. 6 Dimensions [mm]

4.1.2 Selecting the Place of Installation

The EtherCAT bus node is a decentralized component.

- ➔ Use the criteria below to select the place of installation directly next to the machine:
- Short line paths to all components.
 - Provide flat fitting surface to achieve mechanically stress-free mounting.
 - Device diagnostic LEDs visible.
 - Sufficient space to replace devices easily and for making connections and hooking up connecting wires.
 - Protected to prevent ripping out connection lines.
 - Suitable to avoid vibration and shock loads, temperature and humidity (see Section 10.1 "Technical Data").

4.1.3 Fitting the Device

CAUTION

Risk of material damage

Connections may be damaged by improper tampering.

- Protect the device from tampering.

CAUTION

Risk of material damage

The fixing screws are dependent on the substrate at the place of installation.

- ➔ Use the right screws suitable for the type of substrate.

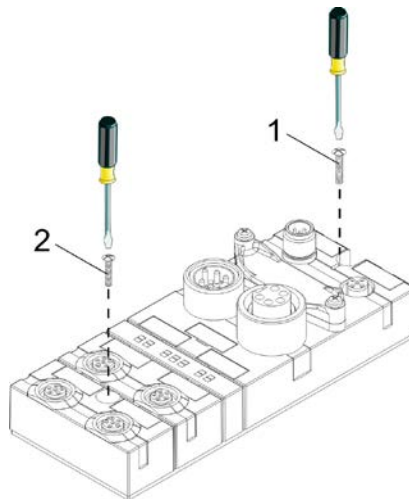


Fig. 7 Mounting

Legend

- | | |
|---|-----------------------------------|
| 1 | Screw M4, washer, grounding strap |
| 2 | Screw M4, washer |



INFORMATION

The illustrated screws, washers and the grounding strap are not part of the scope of supply. The grounding strap is obtainable as an accessory (see Section 7.2.11).

Mount see Fig. 7
(keep to the sequence)

- ➔ (1) Fix the device and grounding strap using an M4 screw and washer.
Tightening torque: 1.2 ± 0.2 Nm, 10.6 ± 1.8 lbf-in.
- ➔ Align the case.
- ➔ (2) Fit the M4 screw and the washer.
Tightening torque: 1.2 ± 0.2 Nm, 10.6 ± 1.8 lbf-in.

4.2 Electrical Installation of EtherCAT



WARNING



Danger from electric voltage in machine.

The result may be death or fatal injuries.

1. Observe the five safety rules.
2. Connect the device to the electrical system.

4.2.1 Conditions

EtherCAT master

The EtherCAT master supports MDP (Modular Device Profile).

Power Supply

Power supplies correspond to SELV or PELV.

Power sources in accordance with EN 61558-2-6 (transformer) or EN 60950-1 (switched-mode power supply) meet these requirements.

Cables

Only use cables that meet the requirements and regulations for safety, electromagnetic compatibility, and, if necessary, telecommunications terminal equipment, and their specifications.

- ➔ Use at least 100base TX with cable type CAT5 (IEEE 802.3u-specified, EIA/TIA-568).
Visit the Accessories section of the Murrelektronik GmbH Online Shop for suitable cables.
- ➔ Ask Murrelektronik GmbH for self-connecting connectors.

4.2.2 Installing



WARNING

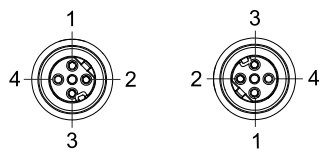


Fire risk from short-circuits

Power supply lines and/or modules damaged by short-circuits may cause overheating and fires.

→ Provide an intelligent power monitoring system or fuse.

M12 sockets for EtherCAT bus



Port 1 and Port 2 are 2 M12 sockets for EtherCAT.

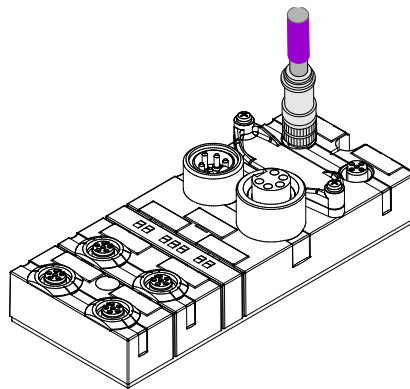


Fig. 8 Sample connection of M12

Connecting

- Remember that the maximum cable length is 100 m to the end point if no auxiliary resources are used. A 2-port switch is integrated in the bus node.
- Use torque wrench Art. No. 7000-99102-000 0000 (see Section 7.5.3).
- Use suitable cables to interconnect the modules and connect them to the EtherCAT master.
Tightening torque: $0.6^{+0.1}$ Nm, $5.3^{+0.9}$ lbf-in.

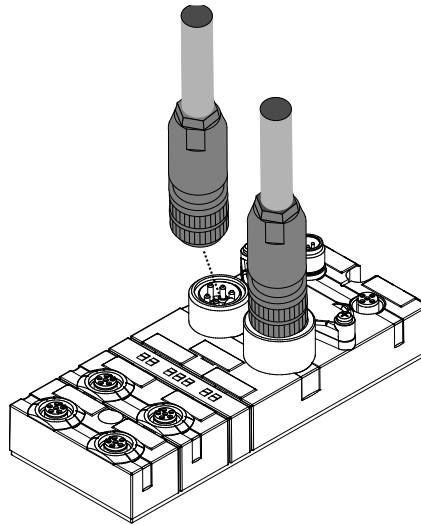
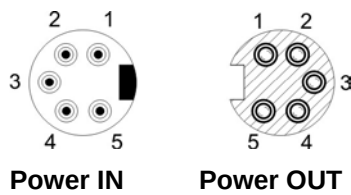
POWER IN, POWER OUT**7/8" connector**

Fig. 9 Sample connection of 7/8"

Connecting

- ➔ Use torque wrench Art. No. 7000-99104-000 0000 (see Section 7.5.3).
- ➔ Plug the 7/8" connector to the 7/8" connectors of the lines.
Tightening torque: 1.5 ± 0.2 Nm, 13.27 ± 1.8 lbf-in.

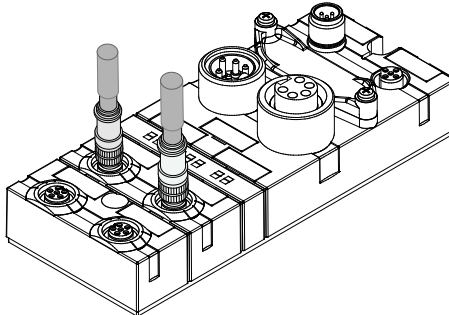
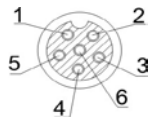
M12 sockets for the Cube67 System

Fig. 10 Sample connection of M12

Connecting

- ➔ Use tool Art. No. 7000-99102-000 0000 (see Section 7.5.3).
- ➔ Connect the M12 sockets to the plugs of the connecting lines.
Tightening torque: 0.6 ± 0.1 Nm, 5.3 ± 0.9 lbf-in.

4.3 Ensuring Tightness (IP67)

CAUTION

Material damage caused by plugs and sockets that are not properly connected.

Protection Class IP67 is only ensured if all plug or screw plug connections are properly closed.

→ Close unused plugs and sockets carefully.

**POWER IN, POWER OUT
7/8" connector**

→ Insert a 7/8" screw plug in an unused slot (see Section 7.5.2).

**M12 sockets for EtherCAT
bus**

→ Insert an M12 screw plug in an unused slot (see Section 7.5.2).

**M12 sockets
for the Cube67 System**

→ Insert a terminal resistor in unused sockets (0,1,2,3) (see Section 7.5.2).

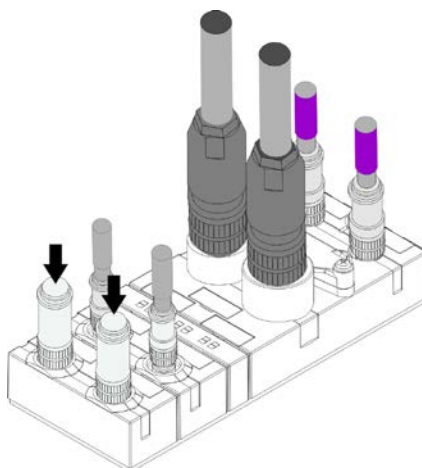


Fig. 11 Example with 2 terminal resistors



INFORMATION

Screw plugs and terminal resistor are obtainable as accessories (see Section 7.5.2).

4.4 Terminating the Internal System Connection

Ensure data transmission

→ At the start and end of each segment of the internal system connection, **always** connect a terminal resistor, even if no modules are connected.

5 Startup

5.1 Requirements

Minimum configuration	An EtherCAT network comprises at least the following components: • one EtherCAT master • one or several slave users • Ethernet cables and plugs to connect the users
System data	To operate the Cube67+ BN-EC, you require the ESI file • Murrelektronik-56527_Basis.xml and files in the Murrelektronik_Cube67 subdirectory: • Murrelektronik-56527_DIO.xml • Murrelektronik-56527_DI.xml • Murrelektronik-56527_DO.xml • Murrelektronik-56527_DI_DO.xml • Murrelektronik-56527_AI.xml • Murrelektronik-56527_AO.xml • Murrelektronik-56527_Func.xml • Murrelektronik-56527_Cube20.xml • Murrelektronik-56527_Branch.xml ➔ Download the ZIP file containing the ESI files from the Murrelektronik website (www.murrelektronik.com)
Rotary switch	The rotary switches are used to set the EtherCAT device ID. • The range is between 000 ... 999. ➔ After the address has been set, the address cover must be put on again and both screws must be tightened with the torque of 0.3 Nm.

**NOTE**

Delivery state: The rotary switches are set to 000.

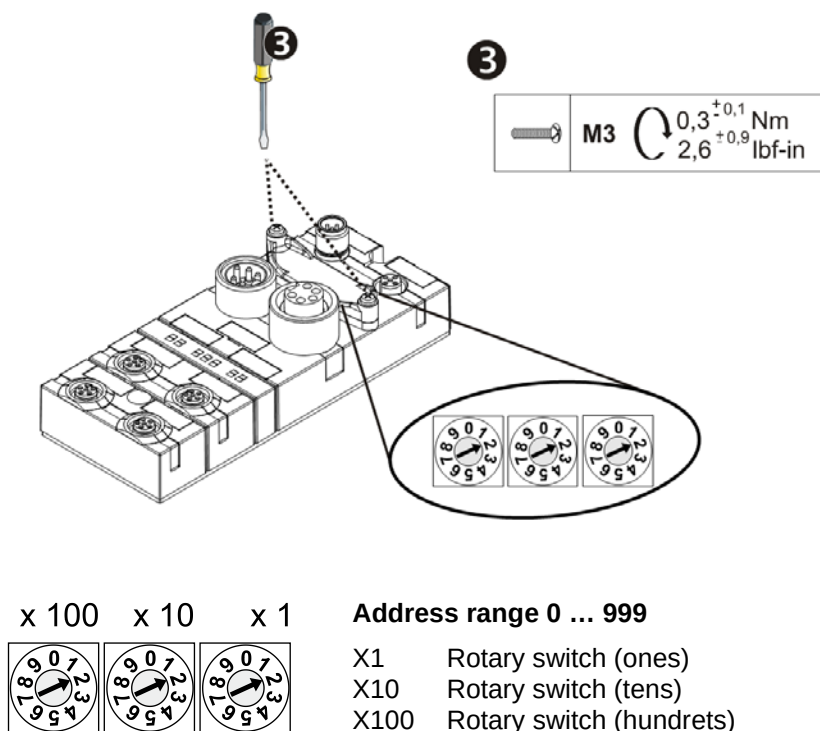


Fig. 12 Rotary switch and addressing BN-EC Cube67+

5.2 ESI Files

Install ESI files

- ➔ Refer to the User Manual of the EtherCAT master where you should store the ESI files.
- ➔ Copy the ZIP file to the specified location.
- ➔ Extract the files from the ZIP file.

The folder hierarchy is created while the files are extracted.

Install ESI files with Twin-CAT®

Refer to Sections 5.4 and 5.5 for setup instructions using TwinCAT.

5.3 General Configuration Notes



INFORMATION

The examples are based on TwinCAT System Manager Version V2.11.0 from Beckhoff GmbH. For more details on TwinCAT-related topics, refer to the TwinCAT System Manager.

5.4 Advanced Users: Configuring with TwinCAT

This section

This section contains summarized configuration and startup information for a Cube67+ BN-EC. For more details, please refer to the ***TwinCAT System Manager***.

Download ESI files

- Download the ESI files from the Murrelektronik GmbH website (www.murrelektronik.com), see Section 5.2.

Connecting

- Connect the Cube67+ BN-EC and the expansion modules to the EtherCAT master (see Section 4.2).

Select or create new project

- Refer to the instructions of the software producer.
 - Add the Cube67+ BN-EC to an existing project.
- or
- Create a new project.

Copy ESI files

- Copy the ESI files of the Cube67+ BN-EC to the TwinCAT folder.
Default folder: ***/Programme/TwinCAT/IO/EtherCAT***
The folder can contain additional ESI files.

Add Cube67+ BN-EC

- Open the TwinCAT® System Manager.
 - Create a new project, if necessary.
 - Add Cube67+ BN-EC to the project.
- or
- Scan for I/O devices.

Select EtherCAT slaves

- Right-click on the entry ***Cube67+ BN-EC***
 - Select all connected EtherCAT slaves (boxes) manually.
- or
- Click on Scan. The ***TwinCAT System Manager*** searches automatically for connected EtherCAT slaves.

5.5 Entry-Level Users: Configuring with TwinCAT

This section

This section contains summarized configuration and startup information for a Cube67+ BN-EC. For more details, please refer to the *TwinCAT System Manager*.

Install TwinCAT

- ➔ Install *TwinCAT*.
- ➔ Install the correct *RT Ethernet connection*.

Download and install ESI file

The TwinCAT® EtherCAT Master/System Manager requires the device description files of the Cube67+ BN-EC to create a configuration in online and offline modes. The device descriptions are called ESI (EtherCAT Slave Information) in the form of XML files).

The description of the Cube67+ BN-EC and its possible expansion modules are distributed among several ESI files (see Section 5.1).

- ➔ Download the ESI files from the Murrelektronik GmbH website (www.murrelektronik.com).
- ➔ Copy the ESI files for the Cube67+ BN-EC to the TwinCAT folder in which the other ESI files are stored (default folder: /TwinCAT/IO/EtherCAT)
- ➔ Open a new System Manager window.

TwinCAT loads the files once, if they have changed since the last time you opened a System Manager window.

Connecting

- ➔ Connect the Cube67+ BN-EC and the expansion modules to the controller.

5.5.1 Creating a New Project

Create new project in Twin-CAT System Manager

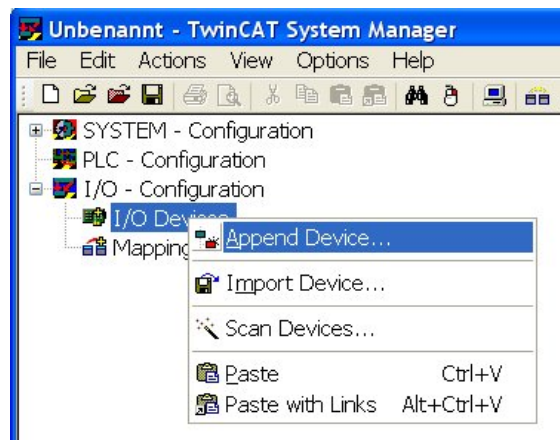


Fig. 13 Inserting a device.

- ➔ Start the **TwinCAT System Manager**.
- ➔ In **I/O – Configuration / I/O Device**, select the item **Append Device** (Fig. 13).

The **Insert Device** window appears.

- ➔ Select from the **EtherCAT / EtherCAT** list.
- ➔ Click **OK** to create a new EtherCAT device.

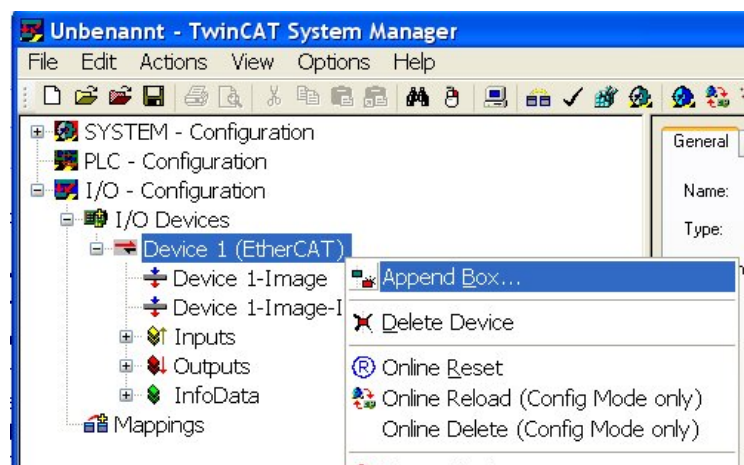


Fig. 14 Function of "Append Box"

- ➔ Click **Append Box** in the Context menu of the inserted EtherCAT device (see Fig. 14).

The **Insert EtherCAT Device** window appears.

- ➔ Insert the **Cube67+ BN-EC** as slave.

Add Cube67+ BN-EC

- ➔ Select the **Box 1 (Cube67+ BN-EC)** option.
- ➔ Select the **Slots** tab.

Line 0 is inserted automatically.

Besides the **Cube67+ BN-EC / Module 1 (56527)**, Line 0 (**Module 2 (Branch 0)**) is also inserted in the configuration automatically after the bus node (Fig. 14).

Line 0 is then automatically inserted in the bus node.

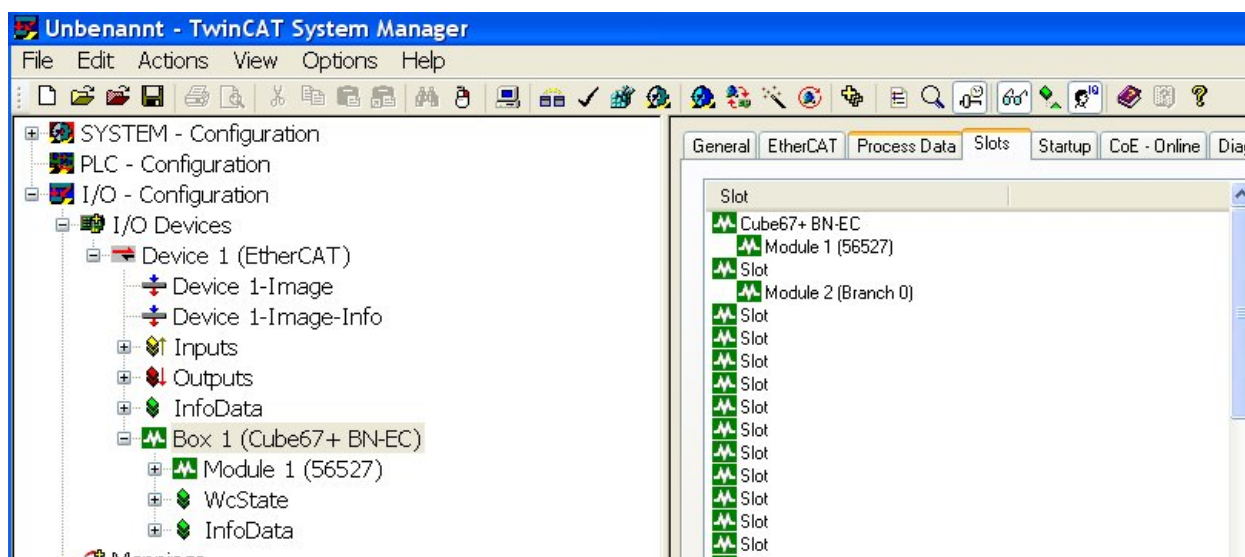


Fig. 15 Configuration after inserting the Cube67+ BN-EC

Insert expansion modules

- ➔ Click on the next free **Slot**.
- ➔ In the right window, select the required expansion modules (see Fig. 15).



NOTE

The virtual line modules **Branch 0**, **Branch 1**, **Branch 2**, and **Branch 3** must always be present in the configuration, even if they contain no expansion modules.

Example

An expansion module Type 56601 DIO16 E 8xM12 at Line 0 (see Fig. 16)

Procedure:

- ➔ Insert the **Cube67+ BN-EC** as slave.
- Branch 2 is inserted automatically.
- ➔ Select the Box 1 (Cube67+ BN-EC) option.
- ➔ Select the **Slots** tab.
- ➔ Click on the next free **Slot**.
- ➔ In the right window, select the **56601 DIO16 E 8xM12** module.
- ➔ Insert branch modules Branch 1, Branch 2, and Branch 3 from Cube67.

Module

1

2

3

4, 5, 6

Save configuration

- ➔ Click on **File** in the menu bar.
- ➔ Select **Create Project Specific XML File ...** to save the configuration.

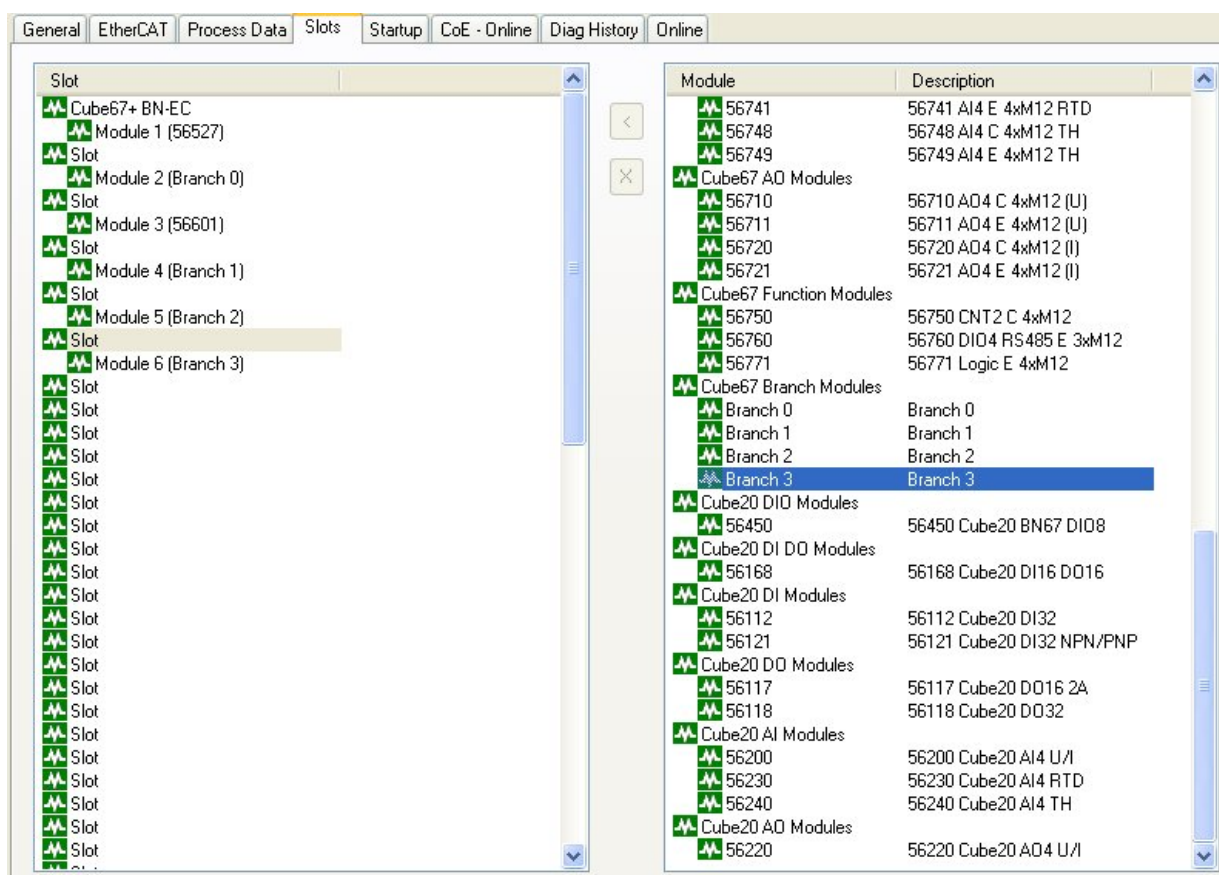


Fig. 16 Sample configuration

5.5.2 Loading An Existing Setup

Scan function in Config Mode

The TwinCAT System Manager reads existing setups using the **Scan** function in **Config Mode**.



Fig. 17 Changing to Config Mode.

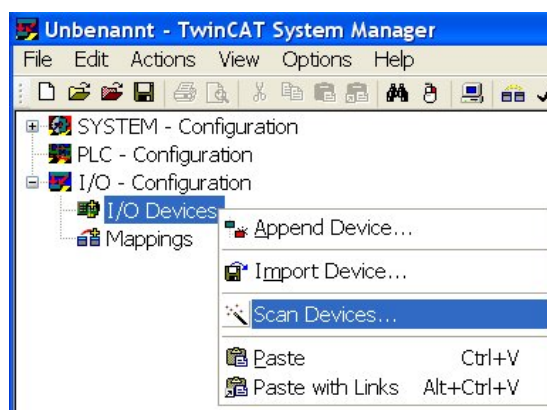


Fig. 18 Scanning device ...

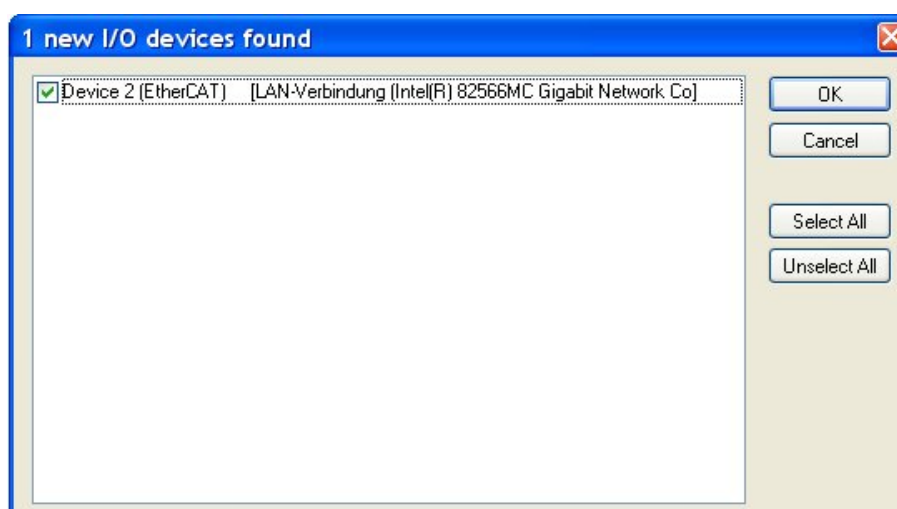


Fig. 19 Selecting the connection to the EtherCAT slave

Scanning

- ➔ Press **Shift-F4** to change to **Config Mode** (see Fig. 17).
- ➔ Select **I/O-Devices/Scan Devices...** (see Fig. 18).
- ➔ In the next window that appears, activate the LAN connection(s) to the EtherCAT slave or EtherCAT slaves. Then click on **OK** (Fig. 19).
- ➔ Click on **Yes** in the **Scan for Boxes** dialog box.

TwinCAT scans the system for existing expansion modules.

- ➔ Click on **Yes** in the **Activate Free Run Mode** dialog box.

TwinCAT starts in **Free Run Mode**.

Insert further modules

- ➔ Insert additional modules and lines as required (see Section 5.5.1).

Save configuration

- ➔ Click on **File** in the menu bar.
- ➔ Select **Create Project Specific XML File ...** to save the configuration.

5.5.2.1 Configuring Outputs During Runtime

Configure in CoE Online, Object 0x40xx

- ➔ Configure the outputs in the **CoE-Online** tab, **Object 0x40xx** (see Fig. 20).
- ➔ Open the parameter(s) by double-clicking.
- ➔ Set the parameter(s) to output (see Fig. 21).

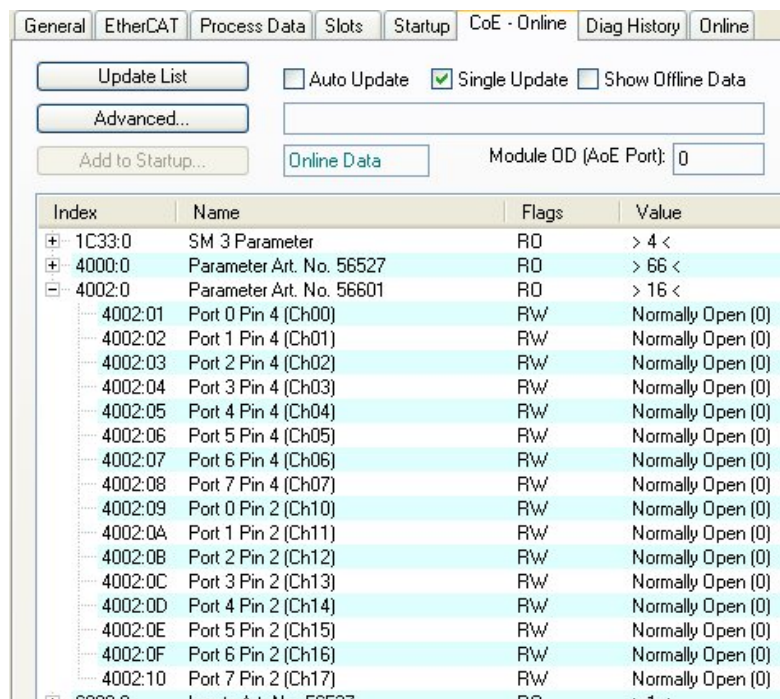


Fig. 20 Configuring the outputs

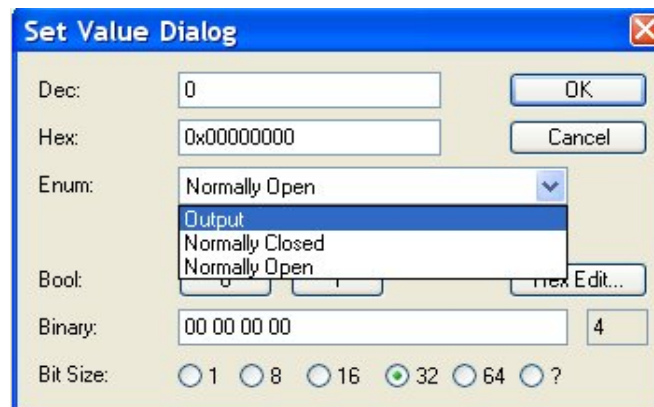


Fig. 21 Setting the parameter to output

Send parameter to bus node → Then send the parameter to the bus node by pressing **F4 (Re-load Devices)**.



NOTE

After saving the configuration and restarting the bus node, the output parameter does **not** remain set.

→ Reconfigure the output parameter.

Output in I/O Overview

An output parameter can also be set in the I/O overview (see Fig. 22).

- ➔ Right-click on the required port in **Output Data**.
- ➔ Select **Online Write** in the Context menu.

The output is called and the output parameter is set.

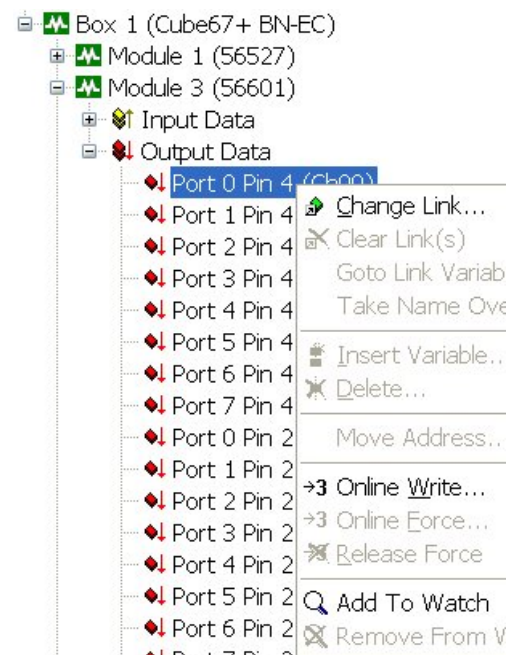


Fig. 22 Setting the parameter to output

5.5.2.2 Configure Outputs on Startup

Configure outputs as PS transitions at startup

You can set configurations as **PS transitions** so that they are valid on startup. Configuration changes are assumed when the status transits from Pre-Operational to Safe-Operational.

Example for an output

- ➔ In the **Startup** tab, open Object **0x40xx** (see Fig. 23).

General EtherCAT Process Data Slots Startup CoE - Online Diag History Online				
Transition	Protocol	Index	Data	Comment
PS	CoE	0x8002:20	0x0200 (512)	Type
PS	CoE	0x4002:01	0x00 (0)	Port 0 Pin 4 (Ch00)
PS	CoE	0x4002:02	0x00 (0)	Port 1 Pin 4 (Ch01)
PS	CoE	0x4002:03	0x00 (0)	Port 2 Pin 4 (Ch02)
PS	CoE	0x4002:04	0x00 (0)	Port 3 Pin 4 (Ch03)
PS	CoE	0x4002:05	0x00 (0)	Port 4 Pin 4 (Ch04)
PS	CoE	0x4002:06	0x00 (0)	Port 5 Pin 4 (Ch05)
PS	CoE	0x4002:07	0x00 (0)	Port 6 Pin 4 (Ch06)
PS	CoE	0x4002:08	0x00 (0)	Port 7 Pin 4 (Ch07)
PS	CoE	0x4002:09	0x00 (0)	Port 0 Pin 2 (Ch10)
PS	CoE	0x4002:0A	0x00 (0)	Port 1 Pin 2 (Ch11)
PS	CoE	0x4002:0B	0x00 (0)	Port 2 Pin 2 (Ch12)
PS	CoE	0x4002:0C	0x00 (0)	Port 3 Pin 2 (Ch13)
PS	CoE	0x4002:0D	0x00 (0)	Port 4 Pin 2 (Ch14)
PS	CoE	0x4002:0E	0x00 (0)	Port 5 Pin 2 (Ch15)
PS	CoE	0x4002:0F	0x00 (0)	Port 6 Pin 2 (Ch16)
PS	CoE	0x4002:10	0x00 (0)	Port 7 Pin 2 (Ch17)

Fig. 23 Example of a startup for a 56601 module

- ➔ Double-click to open the parameter.
- ➔ Set the parameters to the output parameter; e.g. to parameter 2 in the example (see Fig. 24).

The dialog box 'Edit CANopen Startup Entry' contains the following fields and options:

- Transition:** A group box containing six radio buttons:
 - ☐ I → P
 - ☒ P → S
 - ☐ S → P
 - ☐ S → D
 - ☐ D → S
- Index (hex):** A text box containing '4002'.
- Sub-Index (dec):** A text box containing '1'.
- Validate:** An unchecked checkbox.
- Complete Access:** An unchecked checkbox.
- Data (hexbin):** A text box containing '02'.
- Validate Mask:** An empty text box.
- Comment:** A text box containing 'Port 0 Pin 4 (Ch00)'.
- Buttons:** 'OK', 'Cancel', and 'Hex Edit...'.

Fig. 24 Example of output parameter setting

Send parameter to bus node

- ➔ Then send the parameter to the bus node by pressing **F4 (Reload Devices)**.
- ➔ Click on **File** in the menu bar.
- ➔ Save the configuration.

**NOTE**

After saving the configuration and restarting the bus node, the output parameter remains set.

Setting an output in the I/O Overview

An output parameter can also be set in the I/O overview (see Fig. 25).

- ➔ Right-click on the required port in **Output Data**.
- ➔ Select **Online Write** in the Context menu.

The output is called and the output parameter is set.

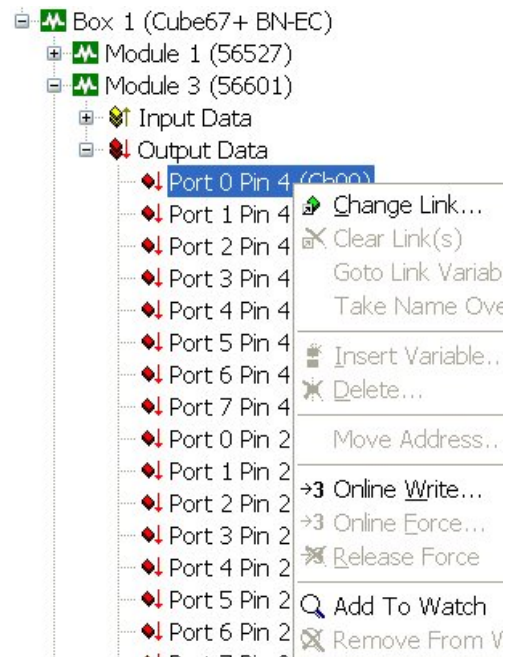


Fig. 25 Setting the parameter to output

5.6 Parameters

5.6.1.1 Parameter Bytes

Byte	Meaning	Definition
0	Global diagnostic reports	Diagnoses are reported or not reported. If you set "Do not report", diagnoses from the expansion modules are also not reported.
	Diagnosis: undervoltage UA at bus node	An undervoltage at UA is reported or not reported.
	Diagnosis: no voltage UA at bus node	"no actuator power supply" is reported or not reported by UA.
	Diagnosis: undervoltage US at bus node	An undervoltage at US is reported or not reported.
1	Reserved	
2	Diagnosis: actuator power supply Slots 3 to 8	Actuator power supply diagnoses, such as "undervoltage" or "no voltage", are reported or not reported for each assigned slot.
3	Diagnosis: actuator power supply Slots 9 to 16	
4	Diagnosis: actuator power supply Slots 17 to 24	
5	Diagnosis: actuator power supply Slots 25 to 32	
6	Diagnosis: actuator power supply Slots 33 to 40	
7	Diagnosis: actuator power supply Slots 41 to 48	
8	Diagnosis: actuator power supply Slots 49 to 56	
9	Diagnosis: actuator power supply Slots 57 to 64	

5.6.1.2 Parameter Overview

Index	Parameters / Diagnoses	Value	Description
0x40yy:01	Global diagnosis	0	Enable
		1	Disable
0x40yy:02	Undervoltage UA (bus node)	0	Signal
		1	Do not signal
0x40yy:03	No voltage (UA) (bus node)	0	Signal
		1	Do not signal
0x40yy:04	Undervoltage US (bus node)	0	Signal
		1	Do not signal
0x40yy:05	Actuator power supply Slot 3	0	Signal
		1	Do not signal
0x40yy:06	Actuator power supply Slot 4	Same as Subindex (SI) 05	
0x40yy:07	Actuator power supply Slot 5	Same as SI 05	
0x40yy:08	Actuator power supply Slot 6	Same as SI 05	
0x40yy:09	Actuator power supply Slot 7	Same as SI 05	
0x40yy:0A	Actuator power supply Slot 8	Same as SI 05	
0x40yy:0B	Actuator power supply Slot 9	Same as SI 05	
0x40yy:0C	Actuator power supply Slot 10	Same as SI 05	
0x40yy:0D	Actuator power supply Slot 11	Same as SI 05	
0x40yy:0E	Actuator power supply Slot 12	Same as SI 05	
0x40yy:0F	Actuator power supply Slot 13	Same as SI 05	
0x40yy:10	Actuator power supply Slot 14	Same as SI 05	

Index	Parameters / Diagnoses	Value	Description
0x40yy:11	Actuator power supply Slot 15	Same as SI 05	
0x40yy:12	Actuator power supply Slot 16	Same as SI 05	
0x40yy:13	Actuator power supply Slot 17	Same as SI 05	
0x40yy:14	Actuator power supply Slot 18	Same as SI 05	
0x40yy:15	Actuator power supply Slot 19	Same as SI 05	
0x40yy:16	Actuator power supply Slot 20	Same as SI 05	
0x40yy:17	Actuator power supply Slot 21	Same as SI 05	
0x40yy:18	Actuator power supply Slot 22	Same as SI 05	
0x40yy:19	Actuator power supply Slot 23	Same as SI 05	
0x40yy:1A	Actuator power supply Slot 24	Same as SI 05	
0x40yy:1B	Actuator power supply Slot 25	Same as SI 05	
0x40yy:1C	Actuator power supply Slot 26	Same as SI 05	
0x40yy:1D	Actuator power supply Slot 27	Same as SI 05	
0x40yy:1E	Actuator power supply Slot 28	Same as SI 05	
0x40yy:1F	Actuator power supply Slot 29	Same as SI 05	
0x40yy:20	Actuator power supply Slot 30	Same as SI 05	
0x40yy:21	Actuator power supply Slot 31	Same as SI 05	
0x40yy:22	Actuator power supply Slot 32	Same as SI 05	
0x40yy:23	Actuator power supply Slot 33	Same as SI 05	
0x40yy:24	Actuator power supply Slot 34	Same as SI 05	
0x40yy:25	Actuator power supply Slot 35	Same as SI 05	
0x40yy:26	Actuator power supply Slot 36	Same as SI 05	
0x40yy:27	Actuator power supply Slot 37	Same as SI 05	
0x40yy:28	Actuator power supply Slot 38	Same as SI 05	
0x40yy:29	Actuator power supply Slot 39	Same as SI 05	
0x40yy:2A	Actuator power supply Slot 40	Same as SI 05	
0x40yy:2B	Actuator power supply Slot 41	Same as SI 05	
0x40yy:2C	Actuator power supply Slot 42	Same as SI 05	
0x40yy:2D	Actuator power supply Slot 43	Same as SI 05	
0x40yy:2E	Actuator power supply Slot 44	Same as SI 05	
0x40yy:2F	Actuator power supply Slot 45	Same as SI 05	
0x40yy:30	Actuator power supply Slot 46	Same as SI 05	
0x40yy:31	Actuator power supply Slot 47	Same as SI 05	
0x40yy:32	Actuator power supply Slot 48	Same as SI 05	
0x40yy:33	Actuator power supply Slot 49	Same as SI 05	
0x40yy:34	Actuator power supply Slot 50	Same as SI 05	
0x40yy:35	Actuator power supply Slot 51	Same as SI 05	
0x40yy:36	Actuator power supply Slot 52	Same as SI 05	
0x40yy:37	Actuator power supply Slot 53	Same as SI 05	
0x40yy:38	Actuator power supply Slot 54	Same as SI 05	
0x40yy:39	Actuator power supply Slot 55	Same as SI 05	
0x40yy:3A	Actuator power supply Slot 56	Same as SI 05	
0x40yy:3B	Actuator power supply Slot 57	Same as SI 05	
0x40yy:3C	Actuator power supply Slot 58	Same as SI 05	
0x40yy:3D	Actuator power supply Slot 59	Same as SI 05	
0x40yy:3E	Actuator power supply Slot 60	Same as SI 05	
0x40yy:3F	Actuator power supply Slot 61	Same as SI 05	
0x40yy:40	Actuator power supply Slot 62	Same as SI 05	
0x40yy:41	Actuator power supply Slot 63	Same as SI 05	
0x40yy:42	Actuator power supply Slot 64	Same as SI 05	

Tab. 2: *Parameter overview*

6 Diagnosis

The Cube67+ System detects and signals many errors independently. It signals errors in two ways:

- Diagnosis by LED indicator
- Diagnosis by Ethernet data

6.1 LED Indicators

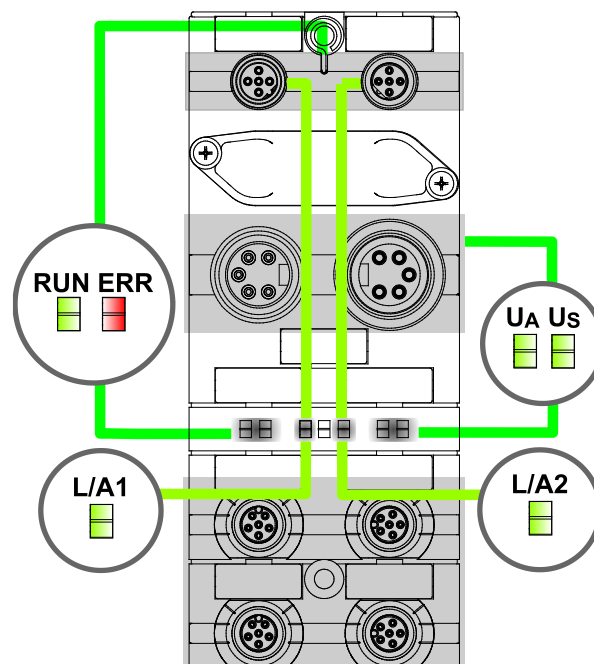


Fig. 26 LED indicators

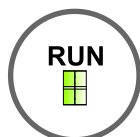
6.1.1 RUN LED States

LED display

The RUN LED indicates the status of the EtherCAT state machine (ESM) in the Cube67+ system.

Color

green



LED Display	LED State	Description
	Lights up continuously	OPERATIONAL status
	Flashes at 5 Hz	PRE-OPERATIONAL status
	Single flash alternating: ON: 200 ms OFF: 1000 ms	SAFE-OPERATIONAL status
	off	OFF or INIT status

Tab. 3: RUN LED display

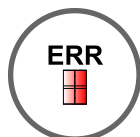
6.1.2 ERR LED States

LED Display

ERR LED indicates system errors.

Color

red



LED Display	LED State	Description
	off	No error detected
	Flashes at 5 Hz	Configuration error
	Single flash alternating: ON: 200 ms OFF: 1000 ms	Status changes due to error, e.g. synchronization error
	Double flash periodic: ON: 200 ms OFF: 200 ms ON: 200 ms OFF: 1000 ms	Watchdog timeout

Tab. 4: ERR LED display

6.1.3 US and UA LED States

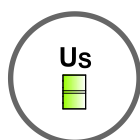
US Sensor and System Power Supply

LED display

The US LED indicates errors in the sensor and system power supply.

Color

green / red



LED Display	LED State	Description
	green permanently lit	OK (> 18 V)
	red permanently lit	Undervoltage
	off	Not available or sensor power supply < 13 V
	red flashing	Internal error

Tab. 5: LED Display US

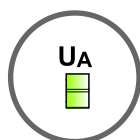
UA actuator power supply

LED display

The UA LED indicates errors in the actuator power supply.

Color

green / red



LED Display	LED State	Description
	green permanently lit	OK (> 18 V)
	red permanently lit	Undervoltage
	off	Not present or actuator power supply < 13 V
	red flashing	Internal error

Tab. 6: UA LED Display

6.1.4 States of US / UA LEDs at Sockets 0 to 3

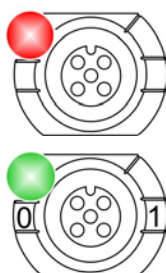
US Status of System Communication

LED display

The US LED indicates the system communication status.

Color

green / red



LED Display	LED State	Description
	green / red permanently lit	Data transfer:
	Green or red flashing	No data exchange
	off	No communication

Tab. 7: LED Display US

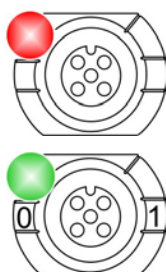
US Sensor and System Power Supply

LED display

The US LED indicates status of the sensor and system power supply.

Color

green / red



LED Display	LED State	Description
	green	OK (> 18 V)
	red	Sensor power supply undervoltage or short-circuit
	red	Overload $I > 4 \text{ A}$
	off	Not available or sensor power supply < 13 V

Tab. 8: LED Display US

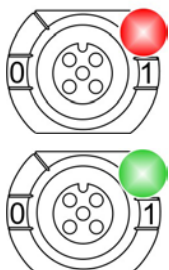
UA Actuator Power Supply

LED display

The UA LED indicates the actuator power supply status.

Color

green / red



LED Display	LED State	Description
	green	OK (> 18 V)
	red	Actuator power supply undervoltage or short-circuit
	red	Overload $I > 4 \text{ A}$
	off	Not available or actuator power supply < 13 A

Tab. 9: UA LED Display

6.1.5 Link/Activity LEDs

L/A1-LED Port 1
L/A2-LED Port 2

Ethernet connection to Port
1

Ethernet connection to Port
2



LED display

The diagnostic LEDs L/A1 and L/A2 indicate the current state of the Ethernet connection.

Color

yellow

LED Display	LED State	Description
	off	Corresponding port is closed, no line connection present.
	Flickering (0 Hz)	Corresponding port is open. Activity detected.
	Permanently ON	Corresponding port is open. No activity detected.

Tab. 10: L/A LED display

6.2 Diagnosis Via The Fieldbus

6.2.1 Diagnosis of Cube67+ BN-EC and I/O Modules



INFORMATION

You will find more details in Document ETG.1020 "EtherCAT Guidelines and Protocol Enhancements"

Diagnosis in EtherCAT

EtherCAT sends diagnoses in separate diagnostic telegrams.

Module and channel diagnoses

EtherCAT supports module and channel diagnoses and sends channel- and module-specific reports.

Storage

Module and channel diagnoses are saved in Object **0x10F3** in subindices **0x06** to **0x85**.

Display

- In the **CoE-Online** tab in the object in hexadecimal notation.
- In the **Diag History** tab, provided this is supported by the Master.

Overwrite Mode

The Cube67+ BN-EC supports **Overwrite Mode**.

You will find details in the EtherCAT Specifications ETG.1020, Section "Diagnosis Handling".

6.2.2 Diagnostic Display in *Diag History* Tab

The **Diag History** tab displays the special error code in the form of a text message.

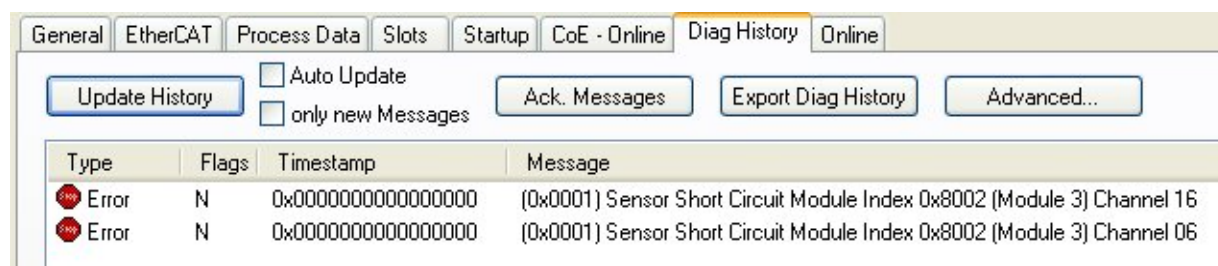


Fig. 27 Diagnostic display in *Diag History*

Type	Flag	Time Stamp	Text Message
 ERROR	N	0x0000000000000000	(0x0001) Sensor shutdown Module 3 Channel 16

Fig. 28 Diagnostic display as text message

6.2.3 Diagnostic Display in CoE-Online Tab

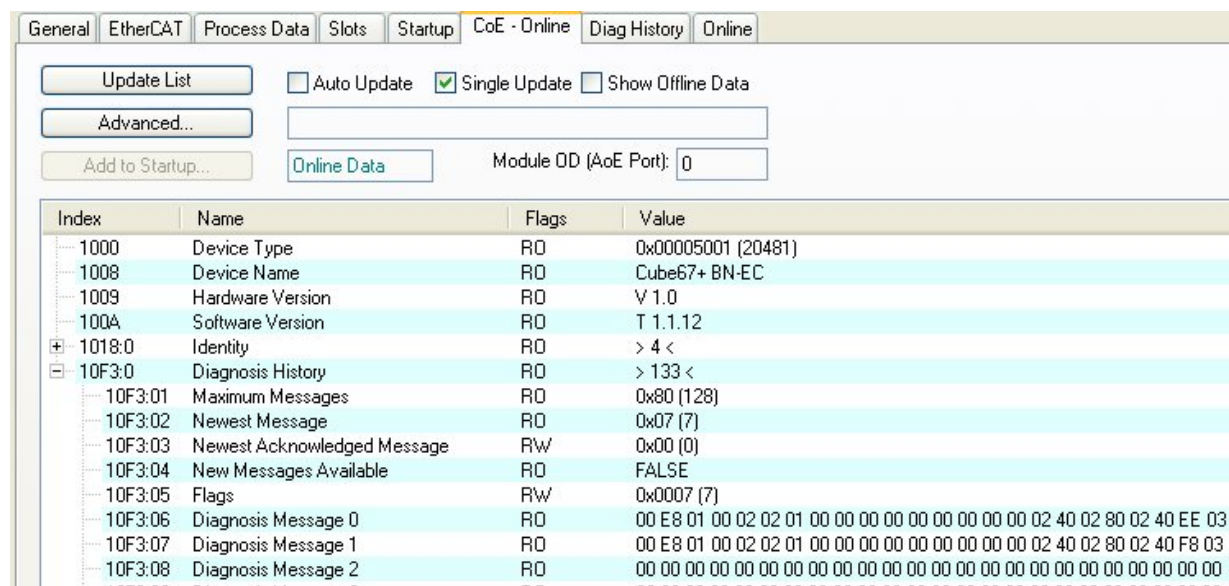


Fig. 29 Diagnostic display in Object 0x10F3

6.2.4 Diagnostic Messages

Diagnostic messages

Diagnostic messages are saved in Object **0x10F3** in subindices **6 to 133**.

The format of the diagnostic messages is described in Section 13.3 in Document ETG.1020.

Length of diagnostic message

The diagnostic message of the Cube67+ BN-EC consists of 24 bytes.

Example

Display of diagnostic message in Object 0x10F3 in *CoE-Online*

Index: 10F3:07

Name: Diagnostic message 1

Flags: RO (read only)

See table below for parameters:

1	2	3	4	5	6	7	8
00 E8 01 00	02 02	01 00	00 00 00 00 00 00 00 00	02 40	02 80	02 40	F8 03

Tab. 11: Example of a diagnostic message

6.2.5 Description of Cube67+ BN-EC Diagnostic Messages

Pos.	Parameters	Description
1	Diag Code Bytes 0 to 3: 00 E8 01 00	Diagnostic code to identify diagnostic message.
		Bits 0-15: 0xE800
		Bits 16-31: Global error code
2	Bytes 4 to 5: 02 02	Bits 0-3: 2 = error message
		Bit 4: 0 = no time indication
		Bits 5-7: Reserved
		Bits 8-15: 2 = number of parameters
3	Text ID Bytes 6 to 7: 01 00	Special error code
4	Time stamp Bytes 8 to 15: 00 00 00 00 00 00 00 00	0 = no time indication
5	Flags parameter 1 Bytes 16 to 17: 02 40	Describes Parameter Type 1
		40: Bits 12-15 = 4: Bits 0-11 contain the size of Parameter 1
		02: Bits 0-11 = 2: size of Parameter 1 in bytes
6	Flags parameter 1 Bytes 18 to 19: 02 80	8002 hex corresponds to Module 3; this module is described in Object 0x8002.
7	Flags parameter 2 Bytes 20 to 21: 02 40	Describes Parameter Type 2
		40: Bits 12-15 = 4: Bits 0-11 contain the size of Parameter 2
		02: Bits 0-11 = 2: size of Parameter 2 in bytes
8	Flags parameter 2 Bytes 22 to 23: F8 03	03F8 hex = 1016 dec: This corresponds to Channel 16.

Tab. 12: Description of diagnostic messages

→ Read the diagnosis as follows:

Pos.	Parameters	Description
1	00 E8	→ Ignore these 2 bytes.
	01 00	→ Reverse the position of the two bytes. Result: 00 01. This hex number corresponds to the global error code as described in Tab. 15: and Tab. 16: . Here 00 01 = short-circuit Another example: 0002 = 0x2 = undervoltage
2	02	Type of message = 2 (errors); the message type always remains the same.
	02	Number of parameters = 2; the number always remains the same.
3	01 00	→ Reverse the position of the two bytes. Result: 00 01. This hex number corresponds to the special error code as described in Tab. 16: . 0001 = 0x1 = sensor short-circuit Another example: 0028 = 0x28 = actuator power supply undervoltage
4	00 00 00 00 00 00 00 00	→ Ignore these 8 bytes. They always remain the same.
5	02 40	→ Ignore these 2 bytes. They always remain the same.
6	02 80	→ Reverse the position of the two bytes. Result: 8002. This corresponds to a hex number (here 0x8002) 0x8002 corresponds to Module 3; this module is described in Object 0x8002. → The module number is obtained by subtracting the hex number 0x7FFF from the object number.
7	02 40	→ Ignore these 2 bytes. They always remain the same.

Pos.	Parameters	Description
8	F8 03	→ Reverse the position of the two bytes. → Result: 03F8. This corresponds to a hex number (here 0x03F8). → Convert 0x03F8 to a decimal number. Result: 2002 dec. → Subtract the following from decimal number 1000: $1016 - 1000 = 16$ This number (16) corresponds to the channel number in which the error occurs.

Tab. 13: *Description of the diagnosis*

6.2.6 EtherCAT Diagnostic Messages

Parameters	Data Type	Use	Description	
Diag Code	UNSIGNED32	M	Diagnostic code to identify diagnostic message.	
			Bits 0-15 = 0x0000-0xDFFF	Not used
			Bits 0-15 = 0xE000-E7FF	Bits 16-31: user definable
			Bits 0-15: 0xE800	Bits 16-31: Emergency Error Code (defined in DS301 or DS4xxx)
			Bits 0-15: 0xE801-0xEFFF	Reserved for later standardization.
			Bits 0-15: 0xF000-0xFFFF	Not used
Flags	Unsigned16	M	Bits 0-3	Diagnostic type
				0 Information
				1 Warning message
				2 Error message
				3-15 Reserved for later standardization.
			Bit 4	Time stamp is a local stamp. The global time stamp can be calculated by reading the current time stamp (Object 0x10F8) and calculating the age of the message.
			Bits 5-7	Reserved for later standardization.
			Bits 8-15	Number of parameters in the diagnostic message
Text ID	Unsigned16	M	Text ID referenced in the ESI file	
			0	No Text ID
			1-65535	Text ID referenced in the ESI file The Text ID should be unique in connection with the Order No., Product No., and Revision Number.
Time Stamp	UNSIGNED64	M	Time stamp in ns (from DC unit). Or local time stamp (Object 0x10F8) if DC units are not supported or are only supported in 32-bit mode.	
			0	No time indication
			<> 0	Time stamp

Parameters	Data Type	Use	Description	
Flags Parameter 1	Unsigned16	o	Describes Parameter Type 1	
			Bits 12-15 = 0	Bit 0-11 is the data type index of the data type of Parameter 1 0x0001: BOOLEAN: ESI Specification %c 0x0002: INTEGER8: ESI Specification %d 0x0003: INTEGER16: ESI Specification %d 0x0004: INTEGER32: ESI Specification %d 0x0005: UNSIGNED8: ESI Specification %u 0x0006: UNSIGNED16: ESI Specification %u 0x0007: UNSIGNED32: ESI Specification %u
			Bits 12-15 = 1	Bit 0-11 contains the byte matrix of Parameter 1 in byte; Parameter 1 is a byte matrix. Can be used for the following data types: OCTET_STRING (Index 0x00A), Specification %s
			Bits 12-15 = 2	The size of the ASCII character string (without suffix 0) in bytes, Parameter 1 is an ASCII character string. Can be used for the following data types: VISIBLE_STRING (Index 0x009), Specification %s

*Use: M = mandatory, O = optional, C = conditional

Tab. 14: *EtherCAT diagnostic messages*

6.2.7 Error Codes

The following **global error codes** are used in Pos.1 of the diagnostic telegram (in decimal notation):

Error Code No.	Meaning
0	Configuration error
1	Short-circuit
2	Undervoltage
3	Overvoltage (not used at present)
4	Overload (not used at present)
5	Overtemperature (not used at present)
6	Line break
7	Upper limit overshoot
8	Upper limit overshoot
9	Other errors, e.g. actuator parameterization error
16	Parameter error

Tab. 15: *Global error codes*

The following **special error codes** are used in Pos.3 of the diagnostic telegram (in decimal notation):

Error Code No.	Meaning
1	Sensor short-circuit
6	Line break
7	Upper limit overshoot
8	Lower limit undershoot
16	Parameter error
21	Reference channel fault
23	Actuator warning
24	Actuator short-circuit
26	DESINA diagnosis
33	Reference channel fault
40	Actuator power supply undervoltage
41	Actuator power supply not available
42	External actuator power supply undervoltage
43	External actuator power supply not available
44	Actuator power supply overvoltage
45	External actuator power supply overvoltage
50	Sensor power supply undervoltage
51	Sensor power supply not available
52	Sensor power supply overvoltage
60	Actuator parameter error
61	Sensor parameter error
96	Configuration error (rupture of internal system connection)
97	Configuration error (missing module, detected during configuration check)
98	Configuration error (incorrect module, detected during configuration check)
99	Configuration error (rupture of internal system connection)

Tab. 16: *Special error codes*

6.3 Emergency Telegrams

Emergency telegrams can be sent when a device-internal error occurs/expires. They appear in the "Logger Output" window.

Here is an example of an emergency telegram of an occurring diagnosis:

Emergency telegrams

Emergency telegrams can be sent when a device-internal error occurs/expires. They appear in the **Logger Output** window. The error code may be related to a specific manufacturer (0x0000 to 0x9FFF) or linked to a specification (0xA000-0xEFFF).

Example

	(65535) 12.06.2012 08:46:2... Box 1 (Cube67+BN-EC)' (1001): CoE – Emergency (Hex: 3120, 04, '02 00 00 01 00')
---	---

Fig. 30 Example of an emergency telegram

6.3.1 Manufacturer-Specific Emergency Telegram

Pos.	Length	Meaning																																
1	2 bytes	Error codes as described in IEC 61800-7-201. Error codes used in Cube67+ BN-EC: <table><tr><td>0x2100</td><td>Sensor short-circuit</td></tr><tr><td>0x2320</td><td>Actuator shutdown (= short-circuit to 0 V)</td></tr><tr><td>0x3100</td><td>No sensor voltage</td></tr><tr><td>0x3110</td><td>Sensor power supply overvoltage</td></tr><tr><td>0x3120</td><td>Sensor power supply undervoltage</td></tr><tr><td>0x3310</td><td>Actuator warning (= short-circuit to 24 V)</td></tr><tr><td>0x3320</td><td>Actuator power supply undervoltage</td></tr><tr><td>0x7060</td><td>Internal system communication interruption</td></tr><tr><td>0x7090</td><td>Configuration error</td></tr><tr><td>0x9000</td><td>External error (DESINA diagnosis)</td></tr><tr><td>0xF000</td><td>No actuator supply</td></tr><tr><td>0xFF10</td><td>Line break</td></tr><tr><td>0xFF11</td><td>Reference channel fault</td></tr><tr><td>0xFF20</td><td>Range overshoot (analog value of upper limit exceeded)</td></tr><tr><td>0xFF30</td><td>Range undershoot (analog value of lower limit undershot)</td></tr><tr><td>0xFF40</td><td>Parameter error</td></tr></table> The example in Fig. 29 shows error code 0x3120 (sensor power supply undervoltage). If the error code is 00 00, an error has expired.	0x2100	Sensor short-circuit	0x2320	Actuator shutdown (= short-circuit to 0 V)	0x3100	No sensor voltage	0x3110	Sensor power supply overvoltage	0x3120	Sensor power supply undervoltage	0x3310	Actuator warning (= short-circuit to 24 V)	0x3320	Actuator power supply undervoltage	0x7060	Internal system communication interruption	0x7090	Configuration error	0x9000	External error (DESINA diagnosis)	0xF000	No actuator supply	0xFF10	Line break	0xFF11	Reference channel fault	0xFF20	Range overshoot (analog value of upper limit exceeded)	0xFF30	Range undershoot (analog value of lower limit undershot)	0xFF40	Parameter error
0x2100	Sensor short-circuit																																	
0x2320	Actuator shutdown (= short-circuit to 0 V)																																	
0x3100	No sensor voltage																																	
0x3110	Sensor power supply overvoltage																																	
0x3120	Sensor power supply undervoltage																																	
0x3310	Actuator warning (= short-circuit to 24 V)																																	
0x3320	Actuator power supply undervoltage																																	
0x7060	Internal system communication interruption																																	
0x7090	Configuration error																																	
0x9000	External error (DESINA diagnosis)																																	
0xF000	No actuator supply																																	
0xFF10	Line break																																	
0xFF11	Reference channel fault																																	
0xFF20	Range overshoot (analog value of upper limit exceeded)																																	
0xFF30	Range undershoot (analog value of lower limit undershot)																																	
0xFF40	Parameter error																																	

Pos.	Length	Meaning
2	1 byte	Error Register Bit 0 Generic error Bit 1 Current error Bit 2 Voltage error Bit 3 Temperature error Bit 4 Communication error Bit 5 Specific error in device profile Bit 6 Reserved Bit 7 Special manufacturer error The Error Register displays which of the various error types occur at a particular time. If the bit is set, the corresponding error is present. In the example in Fig. 29, the value 0x04, i.e. only Bit 2 (voltage error) is set. The number of voltage errors is not contained in this value.
3	5 Byte	Data Byte 0 This value plus 1 produces the module number. Byte 1 Channel number (low) Byte 2 Channel number (high) Bytes 3 and 4 Manufacturing Status Register. Bytes 3 and 4 show which of the various error types occur at a particular time. The error here is specified more precisely than in the Error Register. If the bit is set, the corresponding error is present. Bits in the Manufacturer Status Register Bit 0 Sensor power supply undervoltage Bit 1 Sensor power supply error Bit 2 actuator power supply undervoltage Bit 3 Actuator power supply error Bit 4 sensor power supply short-circuit Bit 5 actuator power supply short-circuit Bit 6 Poor actuator connection (actuator warning) Bit 7 Desina diagnosis Bit 8 Lower / upper limit of analog Bit 9 Parameter error Bit 10 Communication error Bit 11 Other errors Bits 12 to 15 Reserved

Tab. 17: *Format of a manufacturer-specific emergency telegram*

Example Fig. 29

Pos.	Length	Meaning
3	5 Byte	The following values are present in the example Fig. 29: Byte 0 2: Error occurs in Module 3. Byte 1 0 Byte 2 0: Channel number 0 Bytes 3 and 4 0001: Manufacturer Status Register Meaning: Only errors of the sensor power supply undervoltage type are known.

Tab. 18: *Example of a manufacturer-specific emergency telegram*

6.3.2 Specification-Related Emergency Telegram

Specification-related emergency telegrams have the following format:

Pos.	Length	Meaning
1	2 bytes	Error code as described in ETG.1000.6. Error codes used in Cube67+ BN-EC: 0xA000 Status transition from Pre-Operational to Safe-Operational not successful. 0xA001 Status transition from Safe-Operational to Pre-Operational not successful.
2	1 byte	See Specification ETG.1000.6.
3	5 Byte	See Specification ETG.1000.6.

Tab. 19: *Format of a specification-related emergency telegram*

6.4 Troubleshooting

Description of error profiles

This section contains possible error profiles and information that may assist you in troubleshooting.

Error sources at startup

Tab. 20: contains the most frequent error sources on startup. The list does not claim to be complete.

Diagnostic tools

There is a number of useful diagnostic tools from various manufacturers, in particular for checking the correctness of the installation.

6.4.1 Frequent Error Sources at Startup

Error Profile	Diagnosis	LED Display on Bus Node	Possible Cause	Remedial Action
Error on system start	"Incorrect module" error code 97	ERR LED and RUN LED flash.	Configuration error, projected configuration does not correspond with real configuration possibly also cable break in internal system connection	→ Correct the configuration. → Restart.
Error on system start	"Incorrect module" error code 98	ERR LED and RUN LED flash.	Configuration error, projected configuration does not correspond with real configuration possibly also cable break in internal system connection	Correct the configuration. → Restart.
User is not reachable or communication error	"Missing module" error code 96 or error code 99	ERR LED and RUN LED flash.	Cable break, loose contact in internal system connection	→ Rectify cable break or loose contact. → Restart.
User is not reachable.	ESM is "Init" state.	L/A LED are OFF.	No connection between Master and Slave (possibly cable break or loose contact)	→ Rectify cable break or loose contact. → Restart.

Tab. 20: *Frequent error sources at startup*

6.4.2 Status Variable in TWINCAT System Manager (*Online* tab)

Status of Working Counter **WcState**

WcState is a Boolean variable which indicates the working counter status of the coupler.

0: Data are valid.

1: Data are invalid; there is a problem in EtherCAT communication.

Since this command can address more than one slave, the **WcState** variable on all affected devices is set to invalid. For this

reason, it is not possible to detect the device that caused the problem with this variable.

Online Status (InfoData)

InfoData indicates the online status of a single slave.

The slave status can be read out using the **InfoData.State** variable to detect whether a slave has a problem.

InfoData.State

The **State** variable contains the present EtherCAT status of the EtherCAT slave connection.

Value	Description
0x__1	Slave is in INIT status.
0x__2	Slave is in PREOP status.
0x__3	Slave is in BOOT status.
0x__4	Slave is in SAFEOP status.
0x__8	Slave is in OP status.
0x0010	Slave signals an error.
0x0020	Invalid VendorID, read Art. No. ...
0x0100	Slave not present.
0x0200	Slave signals connection error.
0x0400	Slave signals missing connection.
0x0800	Slave signals unexpected connection.
0x1000	Port A is communicating.
0x2000	Port B is communicating.
0x4000	Port C is communicating.
0x8000	Port D is communicating.

Tab. 21: *Special error codes*

ADS address (ADSAddr) (optional)

The **AdsAddr** variable contains the AMS NetID and AMS Port of the slave. It indicates in one line the **AMS netID and the AMS Port** of the slave in a hexadecimal system. This information is required for mailbox communication via **ADS** (Automation Device Specification).

AMS netID and AMS Port are parameters for communication with an EtherCAT slave via **ADS**. The **AdsAddr** variable is added as default if the slave supports a mailbox. The port number is identical to the EtherCAT address of the slave.

7 Annex 1: Supported Modules and Accessories

7.1 Cube67 Digital I/O Modules

7.1.1 Signal Delay of Digital Inputs and Outputs



NOTE

The signal delay of the digital inputs and outputs is dependent on the number of modules connected.

Minimum and maximum signal delays

Number of I/O Modules	Delay [ms]	
	Inputs	Outputs
1	6	5
16	9	12

Tab. 22: Signal Delay of Digital Inputs and Outputs

7.1.2 DIO16 C 8xM12 – Art. No. 56600

7.1.2.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 23: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 24: Output data (PAA)

7.1.2.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 09	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 09	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 09	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 09	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 09	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 09	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 09	

Tab. 25: Parameter overview

7.1.3 DIO16 E 8xM12 – Art. No. 56601

7.1.3.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 26: Input data (PAE)

Output Data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 27: Output data (PAA)

7.1.3.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 09	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 09	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 09	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 09	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 09	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 09	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 09	

Tab. 28: *Parameter overview*

7.1.4 DIO16 C 8xM12 – Art. No. 56602

7.1.4.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 29: Output data (PAA)

7.1.4.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 09	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 09	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 09	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 09	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 09	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 09	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 09	

Tab. 30: *Parameter overview*

7.1.5 DIO16 E 8xM12 – Art. No. 56603

7.1.5.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 31: *Input data (PAE)*

7.1.5.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 09	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 09	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 09	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 09	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 09	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 09	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 09	

Tab. 32: *Parameter overview*

7.1.6 DO12 E 6xM12 K3 – Art. No. 56605

7.1.6.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 33: Input data (PAE)

7.1.6.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.7 DI16 E 8xM12 NPN – Art. No. 56606

7.1.7.1 I/O Data Bit Assignment

Input Data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 34: Input data (PAE)

7.1.7.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	Same as SI 01	
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 01	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 01	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 01	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 01	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 01	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 01	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 01	

Tab. 35: *Parameter overview*

7.1.8 DIO8 C 4xM12 – Art. No. 56610

7.1.8.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 36: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 37: Output data (PAA)

7.1.8.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:06	Port 1 Pin 2 (Ch011)	Same as SI 05	
0x40yy:07	Port 2 Pin 2 (Ch012)	Same as SI 05	
0x40yy:08	Port 3 Pin 2 (Ch013)	Same as SI 05	

Tab. 38: *Parameter overview*

7.1.9 DIO8 E 4xM12 – Art. No. 56611

7.1.9.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 39: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 40: Output data (PAA)

7.1.9.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:06	Port 1 Pin 2 (Ch011)	Same as SI 05	
0x40yy:07	Port 2 Pin 2 (Ch012)	Same as SI 05	
0x40yy:08	Port 3 Pin 2 (Ch013)	Same as SI 05	

Tab. 41: Parameter overview

7.1.10 DI8 C 4xM12 – Art. No. 56612

7.1.10.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 42: Input data (PAE)

7.1.10.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
0x40yy:06	Port 1 Pin 2 (Ch011)	Same as SI 05	
0x40yy:07	Port 2 Pin 2 (Ch012)	Same as SI 05	
0x40yy:08	Port 3 Pin 2 (Ch013)	Same as SI 05	

Tab. 43: *Parameter overview*

7.1.11 DI8 E 4xM12 – Art. No. 56613

7.1.11.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 44: *Input data (PAE)*

7.1.11.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.

**NOTE**

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
0x40yy:06	Port 1 Pin 2 (Ch011)	Same as SI 05	
0x40yy:07	Port 2 Pin 2 (Ch012)	Same as SI 05	
0x40yy:08	Port 3 Pin 2 (Ch013)	Same as SI 05	

Tab. 45: Parameter overview

7.1.12 DI8 E 4xM12 NPN – Art. No. 56616

7.1.12.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 47: Input data (PAE)

7.1.12.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 0 Pin 2 (Ch010)	Same as SI 01	
0x40yy:06	Port 1 Pin 2 (Ch011)	Same as SI 01	
0x40yy:07	Port 2 Pin 2 (Ch012)	Same as SI 01	
0x40yy:08	Port 3 Pin 2 (Ch013)	Same as SI 01	

Tab. 48: *Parameter overview*

7.1.13 DIO8 C 8xM8 – Art. No. 56620

7.1.13.1 I/O Data Bit Assignment

Input Data (PAE)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 49: *Input data (PAE)*

Output Data (PAA)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 50: *Output data (PAA)*

7.1.13.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	

Tab. 51: Parameter overview

7.1.14 DIO8 E 8xM8 – Art. No. 56621

7.1.14.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 52: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 53: Output data (PAA)

7.1.14.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	

Tab. 54: *Parameter overview*

7.1.15 DI8 C 8xM8 – Art. No. 56622

7.1.15.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 55: Input data (PAE)

7.1.15.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	

Tab. 56: Parameter overview

7.1.16 DI8 E 8xM8 – Art. No. 56623

7.1.16.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 57: Input data (PAE)

7.1.16.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	

Tab. 58: Parameter overview

7.1.17 DI8 E 8xM8 NPN – Art. No. 56626

7.1.17.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M8 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 59: Input data (PAE)

7.1.17.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	

Tab. 60: Parameter overview

7.1.18 DIO8 E 4xM12 1A – Art. No. 56631

7.1.18.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 61: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	-	-	-	-	13	12	11	10

Tab. 62: Output data (PAA)

7.1.18.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:06	Port 1 Pin 2 (Ch011)	Same as SI 05	
0x40yy:07	Port 2 Pin 2 (Ch012)	Same as SI 05	
0x40yy:08	Port 3 Pin 2 (Ch013)	Same as SI 05	

Tab. 63: *Parameter overview*

7.1.19 DIO16 C 8xM12 1,6A – Art. No. 56640

7.1.19.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 64: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 65: Output data (PAA)

7.1.19.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 09	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 09	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 09	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 09	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 09	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 09	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 09	

Tab. 66: *Parameter overview*

7.1.20 DIO16 DO16 E 16xM12 (1,6/2A) – Art. No. 56641

7.1.20.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10

Tab. 67: Input data (PAE)

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	17	16	15	14	13	12	11	10
Byte 2 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	27	26	25	24	23	22	21	20
Byte 3 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	37	36	35	34	33	32	31	30

Tab. 68: Output data (PAA)

7.1.20.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the coding of Pin 2 and Pin 4 are not identical.

Parameter Bytes

Byte	Meaning
0	Function of sockets 0 to 3 Pin 4 (Channels 00 to 03)
1	Function of sockets 4 to 7 Pin 4 (Channels 04 to 07)
2	Function of sockets 0 to 3 Pin 2 (Channels 10 to 13)
3	Function of sockets 4 to 7 Pin 2 (Channels 14 to 17)
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 2 Pin 4 (Ch02)	Same as SI 01	
0x40yy:04	Port 3 Pin 4 (Ch03)	Same as SI 01	
0x40yy:05	Port 4 Pin 4 (Ch04)	Same as SI 01	
0x40yy:06	Port 5 Pin 4 (Ch05)	Same as SI 01	
0x40yy:07	Port 6 Pin 4 (Ch06)	Same as SI 01	
0x40yy:08	Port 7 Pin 4 (Ch07)	Same as SI 01	
0x40yy:09	Port 0 Pin 2 (Ch010)	0	Input with NO function
		1	Input with NC function
		2	Diagnostic input
		3	Output
0x40yy:0A	Port 1 Pin 2 (Ch011)	Same as SI 09	
0x40yy:0B	Port 2 Pin 2 (Ch012)	Same as SI 09	
0x40yy:0C	Port 3 Pin 2 (Ch013)	Same as SI 09	
0x40yy:0D	Port 4 Pin 2 (Ch014)	Same as SI 09	
0x40yy:0E	Port 5 Pin 2 (Ch015)	Same as SI 09	
0x40yy:0F	Port 6 Pin 2 (Ch016)	Same as SI 09	
0x40yy:10	Port 7 Pin 2 (Ch017)	Same as SI 09	
0x40yy:11	Actuator voltage UA0	0	Signal
		1	Do not signal
0x40yy:12	Actuator voltage UA1	Same as SI 11	
0x40yy:13	Actuator voltage UA2	Same as SI 11	
0x40yy:14	Actuator voltage UA3	Same as SI 11	

Tab. 69: *Parameter overview*

7.1.21 DO16 C Valve K3 – Art. No. 56650

7.1.21.1 I/O Data Bit Assignment

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08

Tab. 70: Output data (PAA)

7.1.21.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.22 DO16 E Valve – Art. No. 56651

7.1.22.1 I/O Data Bit Assignment

Output data (PAA)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Channels 08 to 15)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08

Tab. 71: Output data (PAA)

7.1.22.2 Parameterization



Note

The parameter bytes of this module are reserved.

7.1.23 DO16 E MAC – Art. No. 56653

7.1.23.1 I/O Data Bit Assignment

Output data (PAA)

Byte 0 (Pin 4 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Pin 2 of M12 sockets)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08

Tab. 72: Output data (PAA)

7.1.23.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.24 DO8 E Valve – Art. No. 56655

7.1.24.1 I/O Data Bit Assignment

Output data (PAA)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 73: Output data (PAA)

7.1.24.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.25 DO32 E Valve 0,5A – Art. No. 56656

7.1.25.1 I/O Data Bit Assignment

Output data (PAA)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Channels 08 to 15)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08
Byte 2 (Channels 16 to 23)								
Bit	7	6	5	4	3	2	1	0
Channel	23	22	21	20	19	18	17	16
Byte 3 (Channels 24 to 31)								
Bit	7	6	5	4	3	2	1	0
Channel	31	30	29	28	27	26	25	24

Tab. 74: Output data (PAA)

7.1.25.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.26 DO32 E MAC – Art. No. 56657

7.1.26.1 I/O Data Bit Assignment

Output data (PAA)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Channels 08 to 15)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08
Byte 2 (Channels 16 to 23)								
Bit	7	6	5	4	3	2	1	0
Channel	23	22	21	20	19	18	17	16
Byte 3 (Channels 24 to 31)								
Bit	7	6	5	4	3	2	1	0
Channel	31	30	29	28	27	26	25	24

Tab. 75: Output data (PAA)

7.1.26.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.27 DIO8 E Cable – Art. No. 56661

7.1.27.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Byte 0)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 76: Input data (PAE)

Output data (PAA)

Byte 0 (Byte 0)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 77: Output data (PAA)

7.1.27.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.

Parameter Bytes

Byte	Meaning
0	Function of Channels 00 to 03
1	Function of Channels 04 to 07
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 01	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Channel 02	Same as Subindex (SI) 01	
0x40yy:03	Channel 03	Same as SI 01	
0x40yy:04	Channel 04	Same as SI 01	
0x40yy:05	Channel 05	Same as SI 01	
0x40yy:06	Channel 06	Same as SI 01	
0x40yy:07	Channel 07	Same as SI 01	
0x40yy:08	Channel 08	Same as SI 01	

Tab. 78: *Parameter overview*

7.1.28 DIO16 E Cable 0,5A – Art. No. 56662

7.1.28.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Channels 08 to 15)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08

Tab. 79: Input data (PAE)

Output data (PAA)

Byte 0 (Byte 0)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Channels 08 to 15)								
Bit	7	6	5	4	3	2	1	0
Channel	15	14	13	12	11	10	09	08

Tab. 80: Output data (PAA)

7.1.28.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.

Parameter Bytes

Byte	Meaning
0	Function of Channels 00 to 03
1	Function of Channels 04 to 07
2	Function of Channels 08 to 11
3	Function of Channels 12 to 15

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 01	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Channel 02	Same as Subindex (SI) 01	
0x40yy:03	Channel 03	Same as SI 01	
0x40yy:04	Channel 04	Same as SI 01	
0x40yy:05	Channel 05	Same as SI 01	
0x40yy:06	Channel 06	Same as SI 01	
0x40yy:07	Channel 07	Same as SI 01	
0x40yy:08	Channel 08	Same as SI 01	
0x40yy:09	Channel 09	Same as SI 01	
0x40yy:0A	Channel 10	Same as SI 01	
0x40yy:0B	Channel 11	Same as SI 01	
0x40yy:0C	Channel 12	Same as SI 01	
0x40yy:0D	Channel 13	Same as SI 01	
0x40yy:0E	Channel 14	Same as SI 01	
0x40yy:0F	Channel 15	Same as SI 01	
0x40yy:10	Channel 16	Same as SI 01	

Tab. 81: *Parameter overview*

7.1.29 DIO8 E M16 0,5A – Art. No. 56663

7.1.29.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 82: Input data (PAE)

Output data (PAA)

Byte 0								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 83: Output data (PAA)

7.1.29.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.

Parameter Bytes

Byte	Meaning
0	Function of Channels 00 to 03
1	Function of Channels 04 to 07
2	Reserved
3	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 01	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Channel 02	Same as Subindex (SI) 01	
0x40yy:03	Channel 03	Same as SI 01	
0x40yy:04	Channel 04	Same as SI 01	
0x40yy:05	Channel 05	Same as SI 01	
0x40yy:06	Channel 06	Same as SI 01	
0x40yy:07	Channel 07	Same as SI 01	
0x40yy:08	Channel 08	Same as SI 01	

Tab. 84: Parameter overview

7.1.30 DIO8 E Cable M12 ID – Art. No. 56665

7.1.30.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 85: *Input data (PAE)*

Output data (PAA)

Byte 0 (Channels 00 to 07)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 86: *Output data (PAA)*

7.1.30.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.31 DI16/DO16 E Cable 0,5A – Art. No. 56671

7.1.31.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0								
Bit	7	6	5	4	3	2	1	0
Channel	IN 8	7	6	5	4	3	2	IN 1
Byte 1 (Channels 08 to 15)								
Bit	7	6	5	4	3	2	1	0
Channel	IN 16	15	14	13	12	11	10	IN 9

Tab. 87: Input data (PAE)

Output data (PAA)

Byte 0								
Bit	7	6	5	4	3	2	1	0
Channel	Out 8	06	05	04	03	02	01	Out 1
Byte 1								
Bit	7	6	5	4	3	2	1	0
Channel	Out 16	06	05	04	03	02	01	Out 9

Tab. 88: Output data (PAA)

7.1.31.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.1.32 DIO8/DI8 E TB Box – Art. No. 56681

7.1.32.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Terminal X1)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Terminal X0)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 89: Input data (PAE)

Output data (PAA)

Byte 0 (Terminal X1)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Terminal X0)								
Bit	7	6	5	4	3	2	1	0
Channel	Reserved							

Tab. 90: Output data (PAA)

7.1.32.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the function coding of Channels 00 to 07 (X0) and Channels 00 to 07 (X1) is different.

Parameter Bytes

Byte	Meaning
0	Function of Channels 00 to 03
1	Function of Channels 04 to 07
2	Function of Channels 00 to 03
3	Function of Channels 04 to 07
4	Reserved
5	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Terminal X1:00 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Terminal X1:01 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Terminal X1:02 (Ch02)	Same as SI 01	
0x40yy:04	Terminal X1:03 (Ch03)	Same as SI 01	
0x40yy:05	Terminal X1:04 (Ch04)	Same as SI 01	
0x40yy:06	Terminal X1:05 (Ch05)	Same as SI 01	
0x40yy:07	Terminal X1:06 (Ch06)	Same as SI 01	
0x40yy:08	Terminal X1:07 (Ch07)	Same as SI 01	
0x40yy:09	Terminal X0:00 (Ch10)	0	Input with NO function
		1	Input with NC function
0x40yy:0A	Terminal X0:01 (Ch11)	Same as SI 09	
0x40yy:0B	Terminal X0:02 (Ch12)	Same as SI 09	
0x40yy:0C	Terminal X0:03 (Ch13)	Same as SI 09	
0x40yy:0D	Terminal X0:04 (Ch14)	Same as SI 09	
0x40yy:0E	Terminal X0:05 (Ch15)	Same as SI 09	
0x40yy:0F	Terminal X0:06 (Ch16)	Same as SI 09	
0x40yy:10	Terminal X0:07 (Ch17)	Same as SI 09	

Tab. 91: *Parameter overview*

7.1.33 DIO8/DI8 E TB Rail – Art. No. 56691

7.1.33.1 I/O Data Bit Assignment

Input data (PAE)

Byte 0 (Terminal X1)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Terminal X0)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00

Tab. 92: Input data (PAE)

Output data (PAA)

Byte 0 (Terminal X1)								
Bit	7	6	5	4	3	2	1	0
Channel	07	06	05	04	03	02	01	00
Byte 1 (Terminal X0)								
Bit	7	6	5	4	3	2	1	0
Channel	Reserved							

Tab. 93: Output data (PAA)

7.1.33.2 Parameterization

Some parameters have more than two selection options. These parameters are each coded by two bits. The values 0_{dec} to 3_{dec} can then be represented.



NOTE

Note that the function coding of Channels 00 to 07 (X0) and Channels 00 to 07 (X1) is different.

Parameter Bytes

Byte	Meaning
0	Function of Channels 00 to 03
1	Function of Channels 04 to 07
2	Function of Channels 00 to 03
3	Function of Channels 04 to 07
4	Reserved
5	Reserved

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Terminal X1:00 (Ch00)	0	Input with NO function
		1	Input with NC function
		2	Output
0x40yy:02	Terminal X1:01 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Terminal X1:02 (Ch02)	Same as SI 01	
0x40yy:04	Terminal X1:03 (Ch03)	Same as SI 01	
0x40yy:05	Terminal X1:04 (Ch04)	Same as SI 01	
0x40yy:06	Terminal X1:05 (Ch05)	Same as SI 01	
0x40yy:07	Terminal X1:06 (Ch06)	Same as SI 01	
0x40yy:08	Terminal X1:07 (Ch07)	Same as SI 01	
0x40yy:09	Terminal X0:00 (Ch10)	0	Input with NO function
		1	Input with NC function
0x40yy:0A	Terminal X0:01 (Ch11)	Same as SI 09	
0x40yy:0B	Terminal X0:02 (Ch12)	Same as SI 09	
0x40yy:0C	Terminal X0:03 (Ch13)	Same as SI 09	
0x40yy:0D	Terminal X0:04 (Ch14)	Same as SI 09	
0x40yy:0E	Terminal X0:05 (Ch15)	Same as SI 09	
0x40yy:0F	Terminal X0:06 (Ch16)	Same as SI 09	
0x40yy:10	Terminal X0:07 (Ch17)	Same as SI 09	

Tab. 94: *Parameter overview*

7.2 Cube67 Analog I/O Modules

7.2.1 Specific Properties of AI4(U) and AI4(I) Modules

7.2.1.1 Smoothing

Device prefilter

The analog signal can be prefiltered in the device to reduce sensitivity to transient fluctuations in measured values, for example.

Measuring time per channel: 2.0 ms.

Total cycle time: measuring time per channel
+ 2.5 ms.

Chart for time-related smoothing behavior

The chart below shows module behavior to a step response in an active channel.

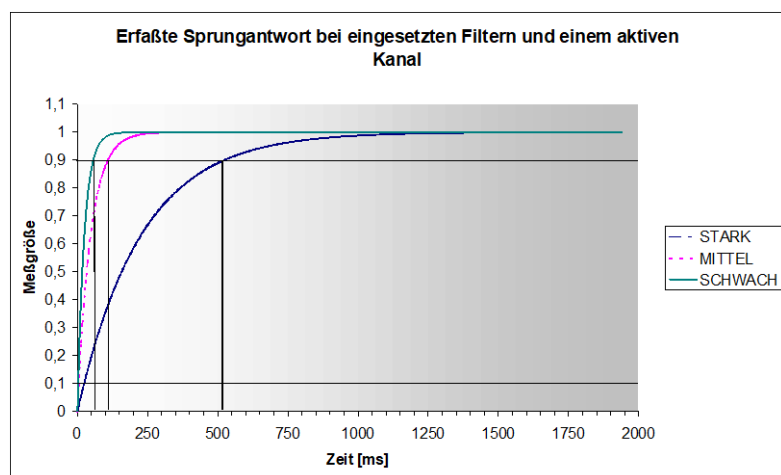


Fig. 31 Smoothing time behavior

7.2.1.2 Delta Value

Define a change in measured value

Use the delta value to define the number of increments by which the measured values must change to the positive or negative in relation to the last measured value so that the analog input value is sent from the AI module to the bus node. This parameter reduces the bus load of the internal system connection.

Example

Last measured value: 1000

Set delta value 100.

A new measured value is sent when it is smaller than 900 or greater than 1100.

7.2.2 Specific Properties of AI4-RTD and AI4-TH Modules

7.2.2.1 50/60 Hz Filter

Use the 50/60 Hz filter to suppress line faults. The module AD converters are then switched to a special mode. As a result, the cycle time is lengthened by about 10 times the default time.



RECOMMENDATION

Always use a filter for Platinum 1000 and Nickel 1000 RTD sensors.

7.2.3 AI4 C 4xM12 (U) – Art. No. 56700

7.2.3.1 I/O Data

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 95: Input data (PAE)

Byte Sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	VZ ¹	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Tab. 96: Byte sequence high/low – Motorola

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	VZ	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸

Tab. 97: Byte sequence low/high– Intel

¹ VZ = prefix

VZ = 0 positive integer

VZ = 1 negative integer

Display of Analog Values

Measuring range 0 to 10 V

Range	Voltage	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>10 V	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	10 V	0111 1111 1111 1111	7FFF	32767	-
	5 V	0011 1111 1111 1111	3FFF	16383	
	0.305 mV	0000 0000 0000 0001	0001	1	
	0 V	0000 0000 0000 0000	0000	0	
Underload	< 0 V	0000 0000 0000 0000	0000	0	-

Tab. 98: Display of analog values, measuring range 0 to 10 V

Measuring range ± 10 V

Range	Voltage	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>10 V	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	10 V	0111 1111 1111 1111	7FFF	32767	-
	5 V	0011 1111 1111 1111	3FFF	16383	
	0.305 mV	0000 0000 0000 0001	0001	1	
	0 V	0000 0000 0000 0000	0000	0	
	-0.305 mV	1111 1111 1111 1111	FFFF	-1	
	-5 V	1100 0000 0000 0000	C000	-16383	
	-10 V	1000 0000 0000 0000	8000	-32768	
Underload	< -10 V	1000 0000 0000 0000	8000	-32768	Yes

Tab. 99: Display of analog values, measuring range ± 10 V

7.2.3.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	Channel 0: delta
3	
4	Settings Channel 1
5	Reserved
6	Channel 1: delta
7	
8	Settings Channel 2
9	Reserved
10	Channel 2: delta
11	
12	Settings Channel 3
13	Reserved
14	Channel 3: delta
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Measuring range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 0: Smoothing	0	No
		1	weak
		2	medium
		3	strong
0x40yy:05	Channel 1: Measuring range	Same as Subindex (SI) 01	
0x40yy:06	Channel 1: Diagnosis	Same as SI 02	
0x40yy:07	Channel 1: Data format	Same as SI 03	
0x40yy:08	Channel 1: Smoothing	Same as SI 04	
0x40yy:09	Channel 2: Measuring range	Same as SI 01	

Index	Parameters	Value	Description
0x40yy:0A	Channel 2: Diagnosis	Same as SI 02	
0x40yy:0B	Channel 2: Data format	Same as SI 03	
0x40yy:0C	Channel 2: Smoothing	Same as SI 04	
0x40yy:0D	Channel 3: Measuring range	Same as SI 01	
0x40yy:0E	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0F	Channel 3: Data format	Same as SI 03	
0x40yy:10	Channel 3: Smoothing	Same as SI 04	

Tab. 100: *Parameter overview***NOTE**

Optimize the cycle time by disabling unused channels.

7.2.4 AI4 E 4xM12 (U) – Art. No. 56701

7.2.4.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 101: *Input data (PAE)*

Byte sequence
High/Low - Motorola (Default Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	VZ ²	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Tab. 102: *Byte sequence high/low – Motorola*

Byte sequence Low/High – Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	VZ	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸

Tab. 103: *Byte sequence low/high– Intel*

² VZ = prefix

VZ = 0 positive integer

VZ = 1 negative integer

Display of Analog Values

Measuring Range 0 to 10 V

Range	voltage	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>10 V	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	10 V	0111 1111 1111 1111	7FFF	32767	-
	5 V	0011 1111 1111 1111	3FFF	16383	
	0.305 mV	0000 0000 0000 0001	0001	1	
	0 V	0000 0000 0000 0000	0000	0	
Underload	< 0 V	0000 0000 0000 0000	0000	0	-

Tab. 104: Display of analog values, measuring range 0 to 10 V

Measuring Range ± 10 V

Range	voltage	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>10 V	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	10 V	0111 1111 1111 1111	7FFF	32767	-
	5 V	0011 1111 1111 1111	3FFF	16383	
	0.305 mV	0000 0000 0000 0001	0001	1	
	0 V	0000 0000 0000 0000	0000	0	
	-0.305 mV	1111 1111 1111 1111	FFFF	-1	
	-5 V	1100 0000 0000 0000	C000	-16383	
	-5 V	1000 0000 0000 0000	8000	-32768	
Underload	< -10 V	1000 0000 0000 0000	8000	-32768	Yes

Tab. 105: Display of analog values, measuring range ± 10 V

7.2.4.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	Channel 0: delta
3	
4	Settings Channel 1
5	Reserved
6	Channel 1: delta
7	
8	Settings Channel 2
9	Reserved
10	Channel 2: delta
11	
12	Settings Channel 3
13	Reserved
14	Channel 3: delta
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Measuring range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 0: Smoothing	0	No
		1	weak
		2	medium
		3	strong
0x40yy:05	Channel 1: Measuring range	Same as Subindex (SI) 01	
0x40yy:06	Channel 1: Diagnosis	Same as SI 02	
0x40yy:07	Channel 1: Data format	Same as SI 03	
0x40yy:08	Channel 1: Smoothing	Same as SI 04	
0x40yy:09	Channel 2: Measuring range	Same as SI 01	
0x40yy:0A	Channel 2: Diagnosis	Same as SI 02	
0x40yy:0B	Channel 2: Data format	Same as SI 03	

Index	Parameters	Value	Description
0x40yy:0C	Channel 2: Smoothing	Same as SI 04	
0x40yy:0D	Channel 3: Measuring range	Same as SI 01	
0x40yy:0E	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0F	Channel 3: Data format	Same as SI 03	
0x40yy:10	Channel 3: Smoothing	Same as SI 04	

Tab. 106: *Parameter overview***NOTE**

Optimize the cycle time by disabling unused channels.

7.2.5 AO4 C 4xM12 (U) – Art. No. 56710

7.2.5.1 I/O Channels

Output data (PAA)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 107: *Input data (PAE)*

Byte sequence High/Low - Motorola (Default Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	VZ ³	-	-	-	-	2 ¹⁰	2 ⁹	2 ⁸

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Tab. 108: *Byte sequence high/low – Motorola*

³ VZ = prefix

VZ = 0 positive integer

VZ = 1 negative integer

Byte sequence Low/High – Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	VZ	-	-	-	-	2^{10}	2^9	2^8

Tab. 109: Byte sequence low/high– Intel

Display of Analog Values

Output Range 0 to 10 V

Range	voltage	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	10 V	0000 0111 1111 1111	07FF	2047
	5 V	0000 0011 1111 1111	03FF	1023
	0.305 mV	0000 0000 0000 0001	0001	1
	0 V	0000 0000 0000 0000	0000	0

Tab. 110: Display of analog values, measuring range 0 to 10 V

Output Range ± 10 V

Range	voltage	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	10 V	0000 0111 1111 1111	07FF	2047
	5 V	0011 1111 1111 1111	03FF	1023
	0.305 mV	0000 0000 0000 0001	0001	1
	0 V	0000 0000 0000 0000	0000	0
	-0.305 mV	1xxx x111 1111 1111	FFFF	-1
	-5 V	1xxx x100 0000 0001	FC00	-1023
	-5 V	1xxx x000 0000 0001	F800	-2048

Tab. 111: Display of analog values, measuring range ± 10 V

7.2.5.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	
3	
4	Settings Channel 1
5	Reserved
6	
7	
8	Settings Channel 2
9	Reserved
10	
11	
12	Settings Channel 3
13	Reserved
14	
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Output range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 1: Output range	Same as Subindex (SI) 01	
0x40yy:05	Channel 1: Diagnosis	Same as SI 02	
0x40yy:06	Channel 1: Data format	Same as SI 03	
0x40yy:07	Channel 2: Output range	Same as SI 01	
0x40yy:08	Channel 2: Diagnosis	Same as SI 02	
0x40yy:09	Channel 2: Data format	Same as SI 03	
0x40yy:0A	Channel 3: Output range	Same as SI 01	
0x40yy:0B	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0C	Channel 3: Data format	Same as SI 03	

Tab. 112: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.2.6 AO4 E 4xM12 (U) – Art. No. 56711

7.2.6.1 I/O Channels

Output data (PAA)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 113: Input data (PAE)

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	VZ ⁴	-	-	-	-	2 ¹⁰	2 ⁹	2 ⁸

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Tab. 114: Byte sequence high/low – Motorola

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	VZ	-	-	-	-	2 ¹⁰	2 ⁹	2 ⁸

Tab. 115: Byte sequence low/high – Intel

⁴ VZ = prefix

VZ = 0 positive integer

VZ = 1 negative integer

Display of Analog Values

Output Range 0 to 10 V

Range	voltage	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	10 V	0000 0111 1111 1111	07FF	2047
	5 V	0000 0011 1111 1111	03FF	1023
	0.305 mV	0000 0000 0000 0001	0001	1
	0 V	0000 0000 0000 0000	0000	0

Tab. 116: Display of analog values, measuring range 0 to 10 V

Output Range ± 10 V

Range	voltage	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	10 V	0000 0111 1111 1111	07FF	2047
	5 V	0011 1111 1111 1111	03FF	1023
	0.305 mV	0000 0000 0000 0001	0001	1
	0 V	0000 0000 0000 0000	0000	0
	-0.305 mV	1xxx x111 1111 1111	FFFF	-1
	-5 V	1xxx x100 0000 0001	FC00	-1023
	-5 V	1xxx x000 0000 0001	F800	-2048

Tab. 117: Display of analog values, measuring range ± 10 V

7.2.6.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	
3	
4	Settings Channel 1
5	Reserved
6	
7	
8	Settings Channel 2
9	Reserved
10	
11	
12	Settings Channel 3
13	Reserved
14	
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Output range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 1: Output range	Same as Subindex (SI) 01	
0x40yy:05	Channel 1: Diagnosis	Same as SI 02	
0x40yy:06	Channel 1: Data format	Same as SI 03	
0x40yy:07	Channel 2: Output range	Same as SI 01	
0x40yy:08	Channel 2: Diagnosis	Same as SI 02	
0x40yy:09	Channel 2: Data format	Same as SI 03	
0x40yy:0A	Channel 3: Output range	Same as SI 01	
0x40yy:0B	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0C	Channel 3: Data format	Same as SI 03	

Tab. 118: Parameter overview

**NOTE**

Optimize the cycle time by disabling unused channels.

7.2.7 AO4 C 4xM12 (I) – Art. No. 56720

7.2.7.1 I/O Channels

Output data (PAA)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 119: Output data (PAA)

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	-	-	-	-	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 120: Byte sequence high/low – Motorola

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	-	-	-	-	2^{10}	2^9	2^8

Tab. 121: Byte sequence low/high – Intel

Display of Analog Values

Output Range 0 to 20 mA

Range	Current [mA]	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	20 mA	0xxx x111 1111 1111	07FF	2047
	10 mA	0xxx x011 1111 1111	03FF	1023
	4 mA	0xxx x001 1001 1001	0199	409
	0 mA	0xxx x000 0000 0000	0000	0

Tab. 122: Display of analog values, output range 0 to 20 mA

Output Range 4 to 20 mA

Range	Current [mA]	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	20 mA	0xxx x111 1111 1111	07FF	2047
	10 mA	0xxx x010 1111 1111	02FF	767
	4 mA	0xxx x000 0000 0000	0000	0

Tab. 123: Display of analog values, output range 4 to 20 mA

7.2.7.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	
3	
4	Settings Channel 1
5	Reserved
6	
7	
8	Settings Channel 2
9	Reserved
10	
11	
12	Settings Channel 3
13	Reserved
14	
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Output range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 1: Output range	Same as Subindex (SI) 01	
0x40yy:05	Channel 1: Diagnosis	Same as SI 02	
0x40yy:06	Channel 1: Data format	Same as SI 03	
0x40yy:07	Channel 2: Output range	Same as SI 01	
0x40yy:08	Channel 2: Diagnosis	Same as SI 02	
0x40yy:09	Channel 2: Data format	Same as SI 03	
0x40yy:0A	Channel 3: Output range	Same as SI 01	
0x40yy:0B	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0C	Channel 3: Data format	Same as SI 03	

Tab. 124: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.2.8 AO4 E 4xM12 (I) – Art. No. 56721

7.2.8.1 I/O Channels

Output data (PAA)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 125: Input data (PAE)

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	-	-	-	-	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 126: Byte sequence high/low – Motorola

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	-	-	-	-	2^{10}	2^9	2^8

Tab. 127: Byte sequence low/high – Intel

Display of Analog Values

Output Range 0 to 20 mA

Range	Current [mA]	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	20 mA	0xxx x111 1111 1111	07FF	2047
	10 mA	0xxx x011 1111 1111	03FF	1023
	4 mA	0xxx x001 1001 1001	0199	409
	0 mA	0xxx x000 0000 0000	0000	0

Tab. 128: Display of analog values, output range 0 to 20 mA

Output Range 4 to 20 mA

Range	Current [mA]	Measured value		
		Binary	Hexadecimal	Decimal
Nominal range	20 mA	0xxx x111 1111 1111	07FF	2047
	10 mA	0xxx x010 1111 1111	02FF	767
	4 mA	0xxx x000 0000 0000	0000	0

Tab. 129: Display of analog values, output range 4 to 20 mA

7.2.8.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	
3	
4	Settings Channel 1
5	Reserved
6	
7	
8	Settings Channel 2
9	Reserved
10	
11	
12	Settings Channel 3
13	Reserved
14	
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Output range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 1: Output range	Same as Subindex (SI) 01	
0x40yy:05	Channel 1: Diagnosis	Same as SI 02	
0x40yy:06	Channel 1: Data format	Same as SI 03	
0x40yy:07	Channel 2: Output range	Same as SI 01	
0x40yy:08	Channel 2: Diagnosis	Same as SI 02	
0x40yy:09	Channel 2: Data format	Same as SI 03	
0x40yy:0A	Channel 3: Output range	Same as SI 01	
0x40yy:0B	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0C	Channel 3: Data format	Same as SI 03	

Tab. 130: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.2.9 AI4 C 4xM12 (I) – Art. No. 56730

7.2.9.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 131: *Input data (PAE)*

Byte sequence
High/Low - Motorola (Default Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 132: *Byte sequence high/low – Motorola*

Byte sequence Low/High – Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8

Tab. 133: *Byte sequence low/high– Intel*

Display of Analog Values

Output Range 0 to 20 mA

Range	Current	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>20 mA	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	20 mA	0111 1111 1111 1111	7FFF	32767	-
	10 mA	0011 1111 1111 1111	3FFF	16383	
	4 mA	0001 1001 1001 1001	1999	6553	
	0 mA	0000 0000 0000 0000	0000	0	
Underload	<0 mA	0000 0000 0000 0000	0000	0	-

Tab. 134: Display of analog values, output range 0 to 20 mA

Output Range 4 to 20 mA

Range	Current	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>20 mA	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	20 mA	0111 1111 1111 1111	7FFF	32767	-
	10 mA	0010 1111 1111 1111	2FFA	12282	
	4 mA	0000 0000 0000 0000	0000	0	
Underload	<4 mA	0000 0000 0000 0000	0000	0	Yes
Line break	<2 mA	0000 0000 0000 0000	0000	0	Yes

Tab. 135: Display of analog values, output range 4 to 20 mA

7.2.9.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	Channel 0: delta
3	
4	Settings Channel 1
5	Reserved
6	Channel 1: delta
7	
8	Settings Channel 2
9	Reserved
10	Channel 2: delta
11	
12	Settings Channel 3
13	Reserved
14	Channel 3: delta
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 0: Measuring range	0	disabled
		1	0 to 10 V
		2	±10 V
0x40yy:02	Channel 0: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 0: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 0: Smoothing	0	No
		1	weak
		2	medium
		3	strong
0x40yy:05	Channel 1: Measuring range	Same as Subindex (SI) 01	
0x40yy:06	Channel 1: Diagnosis	Same as SI 02	
0x40yy:07	Channel 1: Data format	Same as SI 03	
0x40yy:08	Channel 1: Smoothing	Same as SI 04	
0x40yy:09	Channel 2: Measuring range	Same as SI 01	
0x40yy:0A	Channel 2: Diagnosis	Same as SI 02	
0x40yy:0B	Channel 2: Data format	Same as SI 03	

Index	Parameters	Value	Description
	mat		
0x40yy:0C	Channel 2: Smoothing	Same as SI 04	
0x40yy:0D	Channel 3: Measuring range	Same as SI 01	
0x40yy:0E	Channel 3: Diagnosis	Same as SI 02	
0x40yy:0F	Channel 3: Data format	Same as SI 03	
0x40yy:10	Channel 3: Smoothing	Same as SI 04	

Tab. 136: *Parameter overview***NOTE**

Optimize the cycle time by disabling unused channels.

7.2.10 AI4 E 4xM12 (I) – Art. No. 56731

7.2.10.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 137: Input data (PAE)

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 138: Byte sequence high/low – Motorola

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8

Tab. 139: Byte sequence low/high – Intel

Display of Analog Values

Output Range 0 to 20 mA

Range	Current	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>20 mA	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	20 mA	0111 1111 1111 1111	7FFF	32767	-
	10 mA	0011 1111 1111 1111	3FFF	16383	
	4 mA	0001 1001 1001 1001	1999	6553	
	0 mA	0000 0000 0000 0000	0000	0	
Underload	<0 mA	0000 0000 0000 0000	0000	0	-

Tab. 140: Display of analog values, output range 0 to 20 mA

Output Range 4 to 20 mA

Range	Current	Measured value			Diagnosis
		Binary	Hexadecimal	Decimal	
Overload	>20 mA	0111 1111 1111 1111	7FFF	32767	Yes
Nominal range	20 mA	0111 1111 1111 1111	7FFF	32767	-
	10 mA	0010 1111 1111 1111	2FFA	12282	
	4 mA	0000 0000 0000 0000	0000	0	
Underload	<4 mA	0000 0000 0000 0000	0000	0	Yes
Line break	<2 mA	0000 0000 0000 0000	0000	0	Yes

Tab. 141: Display of analog values, output range 4 to 20 mA



NOTE

The overload/underload range is variable.

- When the set overload limits are reached, 0x7FFF is sent (measured value > 20mA).
- Below the underload limit, a zero is output.
- When the overload limit below 20 mA is reached, the value remains at the overload limit.
- In case of a line break, the final value is output (0x7FFF).

7.2.10.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Settings Channels 0 to 3
1	Settings module diagnosis and byte swapping
2	Select filter Channel 0, setting for smoothing and diagnosis Channel 0
3	Select filter Channel 1, setting for smoothing and diagnosis Channel 1
4	Select filter Channel 2, setting for smoothing and diagnosis Channel 2
5	Select filter Channel 3, setting for smoothing and diagnosis Channel 3
6	Limit for range undershoot for 0 to 20 mA Channel 0
7	Limit for range undershoot for 0 to 20 mA Channel 1
8	Limit for range undershoot for 0 to 20 mA Channel 2
9	Limit for range undershoot for 0 to 20 mA Channel 3
10	Limit for range overshoot Channel 0
11	Limit for range overshoot Channel 1
12	Limit for range overshoot Channel 2
13	Limit for range overshoot Channel 3
14	Limit for range undershoot for 4 to 20 mA Channel 0
15	Limit for range undershoot for 4 to 20 mA Channel 1
16	Limit for range undershoot for 4 to 20 mA Channel 2
17	Limit for range undershoot for 4 to 20 mA Channel 3

Tab. 142: *Parameter bytes*

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 00: Measuring range	0	disabled
		1	4 to 20 mA
		2	0 to 20 mA
0x40yy:02	Channel 01: Measuring range	Same as Subindex (SI) 01	
0x40yy:03	Channel 02: Measuring range	Same as SI 01	
0x40yy:04	Channel 03: Measuring range	Same as SI 01	
0x40yy:05	Module / channel diagnosis	0	Signal
		1	Do not signal
0x40yy:06	Channel 00: module Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:07	Channel 00: Smoothing	00	No
		01	weak
		10	medium
		11	strong
0x40yy:08	Channel 00: Line break diagnosis	0	Signal
		1	Do not signal
0x40yy:09	Channel 00: Range under-shoot diagnosis	0	Signal
		1	Do not signal
0x40yy:0A	Channel 00: Range over-shoot diagnosis	0	Signal
		1	Do not signal
0x40yy:0B	Channel 01: Smoothing	Same as SI 07	
0x40yy:0C	Channel 01: Line break diagnosis	Same as SI 08	
0x40yy:0D	Channel 01: Range under-shoot diagnosis	Same as SI 09	
0x40yy:0E	Channel 01: Range over-shoot diagnosis	Same as SI 0A	
0x40yy:0F	Channel 02: Smoothing	Same as SI 07	
0x40yy:10	Channel 02: Line break diagnosis	Same as SI 08	
0x40yy:11	Channel 02: Range under-shoot diagnosis	Same as SI 09	
0x40yy:12	Channel 02: Range over-shoot diagnosis	Same as SI 0A	
0x40yy:13	Channel 03: Smoothing	Same as SI 07	
0x40yy:14	Channel 03: Line break diagnosis	Same as SI 08	

Index	Parameters	Value	Description
0x40yy:15	Channel 03: Range under-shoot diagnosis	Same as SI 09	
0x40yy:16	Channel 03: Range over-shoot diagnosis	Same as SI 0A	

Tab. 143: *Parameter overview*

Settable Parameters



NOTE

The current value is specified with one decimal place.

Parameter 230 corresponds to 23.0 mA,
parameter 40 corresponds to 4.0 mA,

Index	Parameters	Value	Description
0x40yy:17	Channel 0: Range undershoot parameter	0 to 230	Range undershoot setting, data type, Unsigned 8
0x40yy:18	Channel 1: Range undershoot parameter	Same as SI 17	
0x40yy:19	Channel 2: Range undershoot parameter	Same as SI 17	
0x40yy:1A	Channel 3: Range undershoot parameter	Same as SI 17	
0x40yy:1B	Channel 0: Range overshoot parameter	0 to 230	Range overshoot setting, data type, Unsigned 8
0x40yy:1C	Channel 1: Range overshoot parameter	Same as SI 1B	Range overshoot settings, data type, Unsigned 8
0x40yy:1D	Channel 2: Range overshoot parameter	Same as SI 1B	
0x40yy:1E	Channel 3: Range undershoot parameter	Same as SI 1B	
0x40yy:1F	Channel 0: Range undershoot parameter at 4 to 20 mA	20 to 230	Range undershoot setting at 4 to 20 mA, data type, Unsigned 8
0x40yy:20	Channel 1: Range undershoot parameter at 4 to 20 mA	Same as SI 1F	
0x40yy:21	Channel 2: Range undershoot parameter at 4 to 20 mA	Same as SI 1F	
0x40yy:22	Channel 3: Range undershoot parameter at 4 to 20 mA	Same as SI 1F	

Tab. 144: *Settings*

NOTE

Optimize the cycle time by disabling unused channels.

Application of Settable Parameters

Operating mode 0 to 20 mA

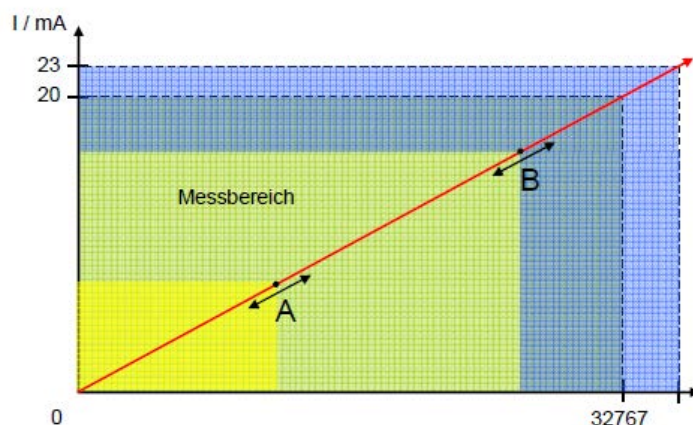


Fig. 32 Settable parameters for range overshoot and undershoot, operating mode 0 to 20 mA

- a** Threshold for range undershoot parameterizable from 0 to 23 mA
- B** Threshold for range overshoot parameterizable from 0 to 23 mA
- A > B** A bus diagnosis takes place⁵ (parameter error) and the diagnostic LED lights up on the affected slot.

⁵ only via Bus Node Art.No. 56527

Operating mode 4 to 20 mA

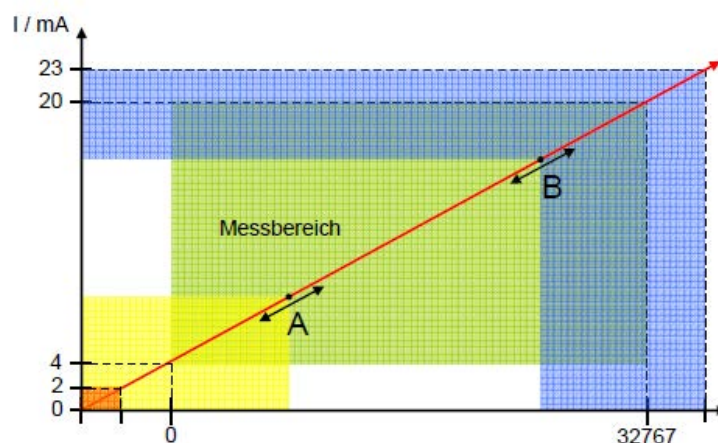


Fig. 33 Settable parameters for range overshoot and undershoot, operating mode 4 to 20 mA

- <2 mA** The module signals line break diagnosis.
- a** Threshold for range undershoot parameterizable from 2 to 23 mA
- B** Threshold for range overshoot parameterizable from 2 to 23 mA
- A > B** A bus diagnosis takes place⁶ (parameter error) and the diagnostic LED lights up on the affected slot.

⁶ only via Bus Node Art.No. 56527

7.2.11 AI4 C 4xM12 RTD – Art. No. 56740

7.2.11.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 145: *Input data (PAE)*

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	-	-	-	-	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 146: *Byte sequence high/low – Motorola*

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	-	-	-	-	2^{10}	2^9	2^8

Tab. 147: *Byte sequence low/high– Intel*

Display of Analog Values

Resistance Measurement

Range	Units		Measuring range 0 to 3000 V	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>7530	>30000	3000 < R < 3251.2	Upper limit
Nominal range	7530	30000	3000	-
	5100	20736	2073.6	
	0	0	0	
Underrun	< 0	< 0	< 0	Yes

Tab. 148: Display of analog values, resistance measurement

Platinum Default

Range 0.1°C	Units		Temperature [°C]	Units		Temperature [°F]	Diagnosis
	Hexa- decimal	Decimal		Hexa- decimal	Decimal		
Overrun	7FFF	32767	-	7FFF	32767	-	Line break
	>2134	>2500	850 < T < 1800	>3D04	>15620	1562 < T < 3272	Upper limit
Nominal range	2134	2500	850	3D04	15620	1562	-
	0	0	0	0	0	0	
	F830	-2000	-200	F330	-3280	-328	
Underrun	<F830	< 2000	< -200	<F330	< -3280	< -328	Yes

Tab. 149: Display of analog values with platinum

Nickel Default

Range 0.1°C	Units		Temperature [°C]	Units		Temperature [°F]	Diagnosis
	Hexa- decimal	Deci- mal		Hexa- decimal	Decimal		
Overrun	7FFF	32767	>1800	7FFF	32767	3272	Line break
	>9C4	>2500	250 < T <1800	>12D4	>4820	482 < T <3272	Upper limit
Nominal range	9C4	2500	250	2134	4820	482	-
	0	0	0	0	0	0	
	FDA8	-600	-60	FD08	-760	-76	
Undershoot	<FDA8	< -600	< -60	<FD08	< -760	< -76	Yes

Tab. 150: *Display of analog values with nickel***7.2.11.2 Parameterization****Parameter Bytes**

Byte	Meaning
0	Global modules settings
1	Metrological settings, Channel 0
2	Metrological settings, Channel 1
3	Metrological settings, Channel 2
4	Metrological settings, Channel 3

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:02	Temperature unit	0	Celsius [°C] (presetting)
		1	Fahrenheit
0x40yy:03	Filter	0	No filter
		1	1 – 50/60 Hz
0x40yy:04	Channel diagnosis	0	On
		1	Off
0x40yy:05	Channel 00: Input	0	On
		1	Off
0x40yy:06	Channel 00: Sensor	0	On
		1	Off
0x40yy:07	Channel 01: Input	Same as SI 05	
0x40yy:08	Channel 01: Sensor	Same as SI 06	
0x40yy:09	Channel 02: Input	Same as SI 05	
0x40yy:0A	Channel 02: Sensor	Same as SI 06	
0x40yy:0B	Channel 03: Input	Same as SI 05	
0x40yy:0C	Channel 03: Sensor	Same as SI 06	

Tab. 151: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.2.12 AI4 E 4xM12 RTD – Art. No. 56741

7.2.12.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 152: *Input data (PAE)*

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	-	-	-	-	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 153: *Byte sequence high/low – Motorola*

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	-	-	-	-	2^{10}	2^9	2^8

Tab. 154: *Byte sequence low/high– Intel*

Display of Analog Values

Resistance Measurement

Range	Units		Measuring range 0 to 3000 V	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	>3251.2	Line break
	>7530	>30000	3000 < R <3251.2	Upper limit
Nominal range	7530	30000	3000	-
	5100	20736	2073,6	
	0	0	0	
Undershoot	< 0	< 0	< 0	Yes

Tab. 155: Display of analog values, resistance measurement

Platinum Default

Range 0.1°C	Units		Temperature [°C]	Units		Temperature [°F]	Diagnosis
	Hexa- decimal	Decimal		Hexa- decimal	Decimal		
Overrun	7FFF	32767	>1800	7FFF	32767	3272	Line break
	>2134	>2500	850 < T <1800	>3D04	>15620	1562 < T <3272	Upper limit
Nominal range	2134	2500	850	3D04	15620	1562	-
	0	0	0	0	0	0	
	F830	-2000	-200	F330	-3280	-328	
Undershoot	<F830	< 2000	< -200	<F330	< -3280	< -328	Yes

Tab. 156: Display of analog values with platinum

Nickel Default

Range 0.1°C	Units		Temperature [°C]	Units		Temperature [°F]	Diagnosis
	Hexa- decimal	Deci- mal		Hexa- decimal	Decimal		
Overrun	7FFF	32767	>1800	7FFF	32767	3272	Line break
	>9C4	>2500	250 < T <1800	>12D4	>4820	482 < T <3272	Upper limit
Nominal range	9C4	2500	250	2134	4820	482	-
	0	0	0	0	0	0	
	FDA8	-600	-60	FD08	-760	-76	
Undershoot	<FDA8	< -600	< -60	<FD08	< -760	< -76	Yes

Tab. 157: *Display of analog values with nickel***7.2.12.2 Parameterization****Parameter Bytes**

Byte	Meaning
0	Global modules settings
1	Metrological settings, Channel 0
2	Metrological settings, Channel 1
3	Metrological settings, Channel 2
4	Metrological settings, Channel 3

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:02	Temperature unit	0	Celsius [°C] (presetting)
		1	Fahrenheit
0x40yy:03	Filter	0	No filter
		1	1 – 50/60 Hz
0x40yy:04	Channel diagnosis	0	On
		1	Off
0x40yy:05	Channel 00: Input	0	On
		1	Off
0x40yy:06	Channel 00: Sensor	0	On
		1	Off
0x40yy:07	Channel 01: Input	Same as SI 05	
0x40yy:08	Channel 01: Sensor	Same as SI 06	
0x40yy:09	Channel 02: Input	Same as SI 05	
0x40yy:0A	Channel 02: Sensor	Same as SI 06	
0x40yy:0B	Channel 03: Input	Same as SI 05	
0x40yy:0C	Channel 03: Sensor	Same as SI 06	

Tab. 158: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.2.13 AI4 C 4xM12 TH – Art. No. 56748

7.2.13.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 159: *Input data (PAE)*

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	-	-	-	-	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 160: *Byte sequence high/low – Motorola*

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	-	-	-	-	2^{10}	2^9	2^8

Tab. 161: *Byte sequence low/high– Intel*

Display of Analog Values

Unit °C (Celsius)

Sensor Type K

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>3598	>13720	1372.0 < T <2000	Upper limit
Nominal range	3598	13720	1372,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 162: *Sensor Type K / °C*

Sensor Type N

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>32C8	>13000	1300.0 < T <2000	Upper limit
Nominal range	32C8	13000	1300,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 163: *Sensor Type N / °C*

Sensor Type J

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>2EE0	>12000	1200.0 < T < 2000	Upper limit
Nominal range	2EE0	12000	1200,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 164: Sensor Type J / °C

Sensor Type E

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>2710	>10000	1000.0 < T < 2000	Upper limit
Nominal range	2710	10000	1000,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 165: Sensor Type E / °C

Sensor Type R

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>4510	>17680	1768.0 < T < 2000	Upper limit
Nominal range	4510	17680	1768.0	-
	0	0	0	
	FE0C	-500	-50.0	
Undershoot	<FE0C	< -500	< -50.0	Yes

Tab. 166: Sensor Type R / °C

Unit °F (Fahrenheit)

Sensor Type K

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>61B8	>25016	2501.6 < T < 3632	Upper limit
Nominal range	61B8	25016	2501,6	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 167: Sensor Type K / °F

Sensor Type N

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>5CA8	>23720	2372.0 < T < 3632	Upper limit
Nominal range	5CA8	23720	2372,0	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 168: Sensor Type N / °F

Sensor Type J

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>55A0	>21920	2192.0 < T < 3632	Upper limit
Nominal range	55A0	21920	2192,0	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 169:
Sensor Type J / °F

Sensor Type E

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>4790	>18320	1832.0 < T < 3632	Upper limit
Nominal range	4790	18320	1832,0	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 170: Sensor Type E / °F

Sensor Type R

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>7D90	>32144	3214.4 < T < 3632	Upper limit
Nominal range	7D90	32144	3214,4	-
	0	0	0	
	FDBC	-580	-58,0	
Undershoot	<FDBC	< -580	< -58,0	Yes

Tab. 171: Sensor Type R / °F

7.2.13.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Global modules settings
1	Metrological settings, Channel 0
2	Metrological settings, Channel 1
3	Metrological settings, Channel 2
4	Metrological settings, Channel 3

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:02	Temperature unit	0	Celsius [°C] (presetting)
		1	Fahrenheit
0x40yy:03	Filter	0	No filter
		1	1 – 50/60 Hz
0x40yy:04	Channel diagnosis	0	On
		1	Off
0x40yy:05	Channel 00: Sensor	0	On
		1	Off
0x40yy:06	Channel 01: Sensor	Same as SI 05	
0x40yy:07	Channel 02: Sensor	Same as SI 05	
0x40yy:08	Channel 03: Sensor	Same as SI 05	

Tab. 172: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.2.14 AI4 E 4xM12 TH – Art. No. 56749

7.2.14.1 I/O Channels

Input data (PAE)

Word	0		1		2		3	
Bit	0	1	2	3	4	5	6	7
Measured value	Channel 0		Channel 1		Channel 2		Channel 3	

Tab. 173: *Input data (PAE)*

Byte sequence
High/Low - Motorola (De-
fault Setting)

Byte	Byte 0 (High Byte)							
Bit	15	14	13	12	11	10	9	8
Value	0	-	-	-	-	2^{10}	2^9	2^8

Byte	Byte 1 (Low Byte)							
Bit	7	6	5	4	3	2	1	0
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Tab. 174: *Byte sequence high/low – Motorola*

Byte sequence Low/High –
Intel

Byte	Byte 0 (Low Byte)							
Bit	15	14	13	12	11	10	9	8
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Byte	Byte 1 (High Byte)							
Bit	7	6	5	4	3	2	1	0
Value	0	-	-	-	-	2^{10}	2^9	2^8

Tab. 175: *Byte sequence low/high– Intel*

Display of Analog Values

Unit °C (Celsius)

Sensor Type K

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>3598	>13720	1372.0 < T <2000	Upper limit
Nominal range	3598	13720	1372,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 176: *Sensor Type K / °C*

Sensor Type N

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>32C8	>13000	1300.0 < T <2000	Upper limit
Nominal range	32C8	13000	1300,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 177: *Sensor Type N / °C*

Sensor Type J

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>2EE0	>12000	1200.0 < T < 2000	Upper limit
Nominal range	2EE0	12000	1200,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 178: Sensor Type J / °C

Sensor Type E

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>2710	>10000	1000.0 < T < 2000	Upper limit
Nominal range	2710	10000	1000,0	-
	0	0	0	
	F830	-2000	-200,0	
Undershoot	<F830	< 2000	< -200,0	Yes

Tab. 179: Sensor Type E / °C

Sensor Type R

Range 0.1°C	Units		Temperature [°C]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>4510	>17680	1768.0 < T < 2000	Upper limit
Nominal range	4510	17680	1768,0	-
	0	0	0	
	FE0C	-500	-50,0	
Undershoot	<FE0C	< -500	< -50,0	Yes

Tab. 180: Sensor Type R / °C

Unit °F (Fahrenheit)

Sensor Type K

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>61B8	>25016	2501.6 < T < 3632	Upper limit
Nominal range	61B8	25016	2501,6	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 181: Sensor Type K / °F

Sensor Type N

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>5CA8	>23720	2372.0 < T < 3632	Upper limit
Nominal range	5CA8	23720	2372,0	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 182: Sensor Type N / °F

Sensor Type J

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>55A0	>21920	2192.0 < T < 3632	Upper limit
Nominal range	55A0	21920	2192,0	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 183: Sensor Type J / °F

Sensor Type E

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>4790	>18320	1832.0 < T < 3632	Upper limit
Nominal range	4790	18320	1832,0	-
	0	0	0	
	F330	-3280	-328,0	
Undershoot	<F330	< -3280	< -328,0	Yes

Tab. 184: Sensor Type E / °F

Sensor Type R

Range 0.1°C	Units		Temperature [°F]	Diagnosis
	Hexadecimal	Decimal		
Overrun	7FFF	32767	-	Line break
	>7D90	>32144	3214.4 < T < 3632	Upper limit
Nominal range	7D90	32144	3214,4	-
	0	0	0	
	FDBC	-580	-58,0	
Undershoot	<FDBC	< -580	< -58,0	Yes

Tab. 185: Sensor Type R / °F

7.2.14.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Global modules settings
1	Metrological settings, Channel 0
2	Metrological settings, Channel 1
3	Metrological settings, Channel 2
4	Metrological settings, Channel 3

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:02	Temperature unit	0	Celsius [°C] (presetting)
		1	Fahrenheit
0x40yy:03	Filter	0	No filter
		1	1 – 50/60 Hz
0x40yy:04	Channel diagnosis	0	On
		1	Off
0x40yy:05	Channel 00: Sensor	0	On
		1	Off
0x40yy:06	Channel 01: Sensor	Same as SI 05	
0x40yy:07	Channel 02: Sensor	Same as SI 05	
0x40yy:08	Channel 03: Sensor	Same as SI 05	

Tab. 186: *Parameter overview*



NOTE

Optimize the cycle time by disabling unused channels.

7.3 Function Modules

7.3.1 CNT2 C 4xM12 – Art. No. 56750

Cyclical data exchange and services

A protocol added to the module permits data read or write to or from the module in cyclical data exchange, and the execution of services.

- Read out software version
- Alternative access to counter status, comparative value and

parameter

- Change parameter.



NOTE

Find out whether there is a software library available for your controller on the Murrelektronik website at www.murrelektronik.com.

7.3.1.1 I/O Channels



NOTE

You will find the access and meaning of the PAE and PAA bytes 2 to 7 in Section 7.3.1.2 “Communication Driver”.

Input Data (PAE)

Byte	0	1	2	3	4	5	6	7
Measured value	Stat_0	Stat_1	CD32I Status	CD32I Index	CD32I Data3	CD32I Data 2	CD32I Data 1	CD32I Data 0

Tab. 187: *Input data (PAE)*

Bit Assignment of Byte Stat_0 :

Bit	Name	Meaning
0	Status_0	Comparative value Counter 0 reached/not reached Status_0 is set when the comparative value is reached. Status_0 is reset when RESET_0 is set.
1	GATE_0	Status of input Counter 0 Enable or disable GATE_0 counter.
2	UPDOWN_0	Status of count direction Counter 0
3	DO_0	Status of output DO_0 DO_0 indicates the status of the digital output 0
4 to 7	-	Reserved

Tab. 188: *Bit assignment of byte Stat_0 :*

Bit Assignment of Byte Stat_1 :

Bit	Name	Meaning
0	Status_1	Comparative value Counter 1 reached/not reached Status_1 is set when the comparative value is reached. Status_1 is reset when RESET_1 is set.
1	GATE_1	Status of input Counter 1 Enable or disable GATE_1 counter.
2	UPDOWN_1	Status of count direction Counter 1
3	DO_1	Status of output DO_1 DO_1 indicates the status of the digital output 1
4 to 7	-	Reserved

Tab. 189: *Bit assignment of byte Stat_1 :***Output Data (PAA)**

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	CD32O Ctrl	CD32O Index	CD32O Data3	CD32O Data 2	CD32O Data 1	CD32O Data 0

Tab. 190: *Output data (PAA)***Bit Assignment of Byte Ctrl_0 :**

Bit	Name	Meaning
0	GATE_0	Activate counter Gate 0 set, counter can start to count
1	RESET_0	Reset counter 0 to start parameter (0000 _{hex})
2	DO_0	Disable or reset output 0 when 1
3	Up/Down_0	0 – process-controlled, 1 - downward PLC controlled
4 to 7	-	Reserved

Tab. 191: *Bit assignment of byte Ctrl_0 :*

Bit Assignment of Byte Ctrl_1 :

Bit	Name	Meaning
0	GATE_1	Activate counter Gate 1 set, counter can start to count.
1	RESET_1	Reset counter 1 to start parameter (0000 _{hex}).
2	DO_1	Disable or reset output 1 when 1.
3	Up/Down_1	0 – process-controlled, 1 - downward PLC controlled
4 to 7	-	Reserved

Tab. 192: *Bit assignment of byte Ctrl_1 :***Behavior of Gate Signal**

Gate	Controller	Process	Meaning
	0	0	Counter disabled
	1	0	Counter enabled
	1	1	Counter enabled
	0	1	Counter enabled

Tab. 193: *Behavior of gate signal***Behavior of Up/Down Signal**

Up/Down	Controller	Process	Meaning
	0	0	Upward count direction
	1	0	Downward counter direction
	1	1	Downward counter direction
	0	1	Downward counter direction

Tab. 194: *Behavior of up/down signal***NOTE**

The counter and the associated outputs work independently. If only the counter is reset, the output retains its status.

- ➔ After the comparative value is reached, reset the counter and the associated output together.

7.3.1.2 Communication Driver

Access codes

The following access codes are written to the output data CTRL (Byte 2):

Decimal values for control byte

	Write Access	Read Access
Byte	192	128
Double word	224	160

Index for input and output data

The tables below contain the associated index (Byte 3) for the input data and output data:

Index	Access	Description	Value	Description
0	Write	Control	0	
1	Read	Status	0	
2	Read	MAX_WR_ADR	28	
3	Read	MAX_RD_ADR	28	
4	Write	PWD_HIGH	0	
5	Write	PWD_LOW	0	
6	Read	Version	0	SW version
7	Read	Version	1	
8	Read	cWert0	0	Counter 0 counter status
9	Read		0	
10	Read		0	
11	Read		0	
12	Read	cWert1	0	Counter 1 counter status
13	Read		0	
14	Read		0	
15	Read		0	
16	Read / Write	Parameter [0]	0	Global modules settings
17	Read / Write	Parameter [1]	0	Settings of Counter 0
18	Read / Write	Parameter [2]	0	Reserved
19	Read / Write	Parameter [3]	0	Comparative value of Counter 0
20	Read / Write	Parameter [4]	0	
21	Read / Write	Parameter [5]	0	

Index	Access	Description	Value	Description
22	Read / Write	Parameter [6]	0	
23	Read / Write	Parameter [7]	0	Settings of Counter 1
24	Read / Write	Parameter [8]	0	Reserved
25	Read / Write	Parameter [9]	0	Comparative value of Counter 1
26	Read / Write	Parameter [10]	0	
27	Read / Write	Parameter [11]	0	
28	Read / Write	Parameter [12]	0	

Tab. 195: Index for input and output data

**NOTE**

You will find the parameters indicated in Tab. 195: in Section 7.3.1.3 “Parameterization”.

7.3.1.3 Parameterization

Parameter Bytes

Byte	Meaning
0	Global modules settings
1	Settings of Counter 0
2	Reserved
3	Comparative value of Channel 0
4	Comparative value of Channel 0
5	Comparative value of Channel 0
6	Comparative value of Channel 0
7	Settings of Counter 1
8	Reserved
9	Comparative value of Channel 1
10	Comparative value of Channel 1
11	Comparative value of Channel 1
12	Comparative value of Channel 1

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Intel on ICAN, Motorola on EtherCAT (presetting)
		1	Motorola on ICAN, Intel on EtherCAT
0x40yy:02	Channel diagnosis	0	On (presetting)
		1	Off
0x40yy:03	Counter 0	0	Counter disabled
		1	Event counting Mode 0
		>1	Mode 0
0x40yy:04	Output 0	0	Output can be reset.
		1	Output is not reset.
0x40yy:05	Polarity of Gate 0	0	not inverted
		1	inverted
0x40yy:06	Polarity of Output (DO) 0	0	not inverted
		1	inverted
0x40yy:07	Comparative value of Counter 0	Range 0x80000000 to 0x7FFFFFFF	Presetting of comparative value, reaction dependent on operating mode.
0x40yy:08	Counter 1	Same as Subindex (SI) 03	
0x40yy:09	Output 1	Same as SI 04	
0x40yy:0A	Polarity of Gate 1	Same as SI 05	
0x40yy:0B	Polarity of Output 1	Same as SI 06	
0x40yy:0C	Comparative value of Counter 1	Same as SI 07	

Tab. 196: *Parameter overview*

Example of "Read Access Double Word"

Sample read access

This section describes a sample read access.

Read access to double word counter status

The counter status is a double word.

Write the access code **160_{dec}** (Read access double word) to the output data Byte 2 (CTRL).

Read access to counter status of counter_0

In this example, we access the counter status of Counter_0 (Index 8).

→ Write the value 8 to the output data Byte 3 (Index).

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	Ctrl 160 _{dec}	Index 8 _{dec}	Data3 0	Data 2 0	Data 1 0	Data 0 0

Tab. 197: Read access to counter status of Counter_0

Reply in input data

If the command was executed successfully, the following reply is sent back in the input data.

Assumed counter status: 0x12345678

Byte	0	1	2	3	4	5	6	7
Measured value	Stat_0	Stat_1	Ctrl 160 _{dec}	Index 8 _{dec}	Data3 12	Data 2 34	Data 1 56	Data 0 78

Tab. 198: Read access to counter status 0x12345678

Reset a command

The previous command must be reset before the next command is executed.

→ Enter a zero in Byte 2 (CTRL).

Byte	0	1	2	3	4	5	6	7
Measured value	Stat_0	Stat_1	Ctrl 0	Index 0	Data3 0	Data 2 0	Data 1 0	Data 0 0

Tab. 199: Reset a command

Example "Write Access to Double Word Comparative Value"

Sample write access

This section describes a sample write access.

Write access to double word comparative value

The comparative value is a double word.

- ➔ Write the access code **224_{dec}** "Write access double word" to the output data of Byte 2 (CTRL).

Comparative value of Counter 1

In this example, we access the comparative value of Counter_1 which is Index 25. The value is 0x7FFFFFFF.

- ➔ Write 25_{dec} to Byte 3 (Index)
- ➔ Write the value 0x7FFFFFFF to Bytes 4 to 7.

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	Ctrl 224 _{dec}	Index 25 _{dec}	Data3 7F	Data2 FF	Data1 FF	Data0 FF

Tab. 200: Write access to the comparative value Counter_1

Reset a command

The previous command must be reset before the next command is executed.

- ➔ Enter a zero in Byte 2 (CTRL).

Byte	0	1	2	3	4	5	6	7
Measured value	Stat_0	Stat_1	Ctrl 0	Index 0	Data3 0	Data 2 0	Data 1 0	Data 0 0

Tab. 201: Reset a command

Example "Read/Write Access to Byte Comparative Value"

Write access to byte value

The comparative value is a byte.

- Write the access code **192_{dec}** "Write access byte" to the output data of Byte 2 (CTRL).

Comparative value of Counter 1

In this example, we access the comparative value of Counter_1 which is Index 25.

- Write 25_{dec} to Byte 3 (Index)
- Write the value 0x7FFF to Bytes 4 and 5.

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	Ctrl 192 _{dec}	Index 25 _{dec}	Data3 7F	Data2 FF	Data1 0	Data0 0

Tab. 202: Write access to the byte value of Counter_1

Reset a command

The previous command must be reset before the next command is executed.

- Enter a zero in Byte 2 (CTRL).

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	Ctrl 0	Index 0	Data3 0	Data 2 0	Data 1 0	Data 0 0

Tab. 203: Reset a command

Read access to comparative value

The comparative value is a byte.

- Write the access code **128_{dec}** "Read access byte" to the output data of Byte 2 (CTRL).

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	Ctrl 128 _{dec}	Index 25 _{dec}	Data3 7F	Data2 FF	Data1 0	Data0 0

Tab. 204: Read access to comparative value

Reply in input data

If the command was executed successfully, the following reply is sent back in the input data.

Byte	0	1	2	3	4	5	6	7
Measured value	Stat_0	Stat_1	Ctrl 128 _{dec}	Index 25 _{dec}	Data3 7F	Data2 FF	Data1 0	Data0 0

Tab. 205: Reply in input data

7.3.2 DIO4 RS485 E 3xM12 – Art. No. 56760

7.3.2.1 I/O Channels

PAE Input Data

Byte	0	1	2	3	4	5	6	7
Measured value	Status_0	Status_1	Status_2	Data4	Data3	Data2	Data1	Data0

Tab. 206: PAE input data

Bit assignment Byte 0:
Status_0

Bit	Name	Value	Meaning
0 to 2	Retval	0	Request_Done
		1	Request_Done
		2	Transmit_Done
		3	Delete RX-Buffer_Done
		4	Delete TX-Buffer_Done
		5	Reset_Done
		6	-
		7	-
3	Rx		Character ready to fetch in Rx buffer
4	In00		Input channel 00
5	In01		Input channel 01
6	In10		Input channel 10
7	In11		Input channel 11

Tab. 207: Bit assignment Byte 0: Status_0

**Bit assignment Byte 1:
Status_1**

Status	Description	Meaning
0	DataLengthRx (0 to 5)	Length of Rx data which are sent in the first transmit PDO. Maximum 5 bytes.
1		
2		
3	ResNb (0 to 31)	Reply number. Incremented every time a PLC request is processed.
4		
5		
6		
7		

Tab. 208: *Bit assignment Byte 1: Status_1*
**Bit assignment Byte 2:
Status_2
(Communication Status)**

Status	Description	Meaning
0	TX_BUSY	Module is sending data.
1	-	-
2	RX_WARN	Rx buffer 80% full
3	RX_PARITY	Parity error on reception
4	XOFF	XOFF was received.
5	TX_OVERFLOW	Tx buffer overflow
6	RX_OVERFLOW	Rx buffer overflow
7	XOFF_TIMEOUT	No XON after XOFF was received within the parameterized monitoring time. The transmit process is resumed.

Tab. 209: *Bit assignment Byte 2: Status_2*

PAA Output Data

Byte	0	1	2	3	4	5	6	7
Measured value	Ctrl_0	Ctrl_1	Ctrl_2	Data4	Data3	Data2	Data1	Data0

Tab. 210: PAA output data

Bit assignment Byte 0: Ctrl_0

Bit	Name	Value	Meaning
0 to 2	CMD	0	Request
		1	Request
		2	Transmit
		3	Delete RX buffer
		4	Delete TX buffer
		5	Reset
		6	-
		7	-
3	-	-	-
4	Out00		Output channel 00
5	Out01		Output channel 01
6	Out10		Output channel 10
7	Out11		Output channel 11

Tab. 211: Bit assignment Byte 0: Ctrl_0

Description of commands

Command	Explanation
Request	At 0->1 or . 1->0, Tx and Rx data are swapped with the buffers.
Transmit	All characters in the Tx buffer are sent. Tx buffer is then empty.
Delete RX buffer	Delete Rx buffer.
Delete TX buffer	Delete Tx buffer. Characters are NOT sent.
Reset	Reset interface; RX and TX buffers are cleared.

Tab. 212: Description of commands

**Bit assignment Byte 1:
Ctrl_1**

Status	Description	Meaning
0	DataLength Tx (0 to 5)	Length of Tx data which are saved to the module Tx buffer. Maximum 5 bytes.
1		
2		
3 to 7	Reserved	

Tab. 213: *Bit assignment Byte 1: Ctrl_1***7.3.2.2 Parameterization****Parameter Bytes**

Byte	Meaning
0	Function of Channel 00,01,10,11 Art. No. 56760
1	Secure state of output 00,01,10,11
2	Settings of RS485 Byte 0
3	Settings of RS485 Byte 1
4	XOFF timeout
5	Send delay

Tab. 214: *Parameter bytes***Parameter Overview**

Index	Parameters	Value	Description
0x40yy:01	Port 0 Pin 4 (Ch00)	0	NC contact
		1	NO contact
		2	Output
0x40yy:02	Port 1 Pin 4 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	Port 0 Pin 2 (Ch010)	0	NC contact
		1	NO contact
		2	Diagnosis input
		3	Output
0x40yy:04	Port 1 Pin 2 (Ch011)	Same as Subindex (SI) 03	

Tab. 215: *Parameter overview*

7.3.3 Logic E 4xM12 – Art. No. 56771

7.3.3.1 I/O Channels

PAE Input Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Channel	Output S4	Output S2	Input socket 2 Pin 4	Input socket 2 Pin 2	Input socket 1 Pin 4	Input socket 1 Pin 2	Input socket 0 Pin 4	Input socket 0 Pin 2

Tab. 216: PAE input data

7.3.3.2 Parameterization

Parameter Bytes

Byte	Meaning
0	Logic settings

Tab. 217: Parameter bytes

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Logic Selection	0	disabled
		1	1x6 AND/NOR
		2	2x3 AND
		3	2x3 XOR

Tab. 218: Parameter overview

7.4 Cube20 Modules



NOTE

Cube20 on the Cube67+ BN-EC can only be used in conjunction with Cube20 BN67 DIO8 module Art. No. 56450 (see Section 7.4.10).

7.4.1 Cube20 DI32 – Art. No. 56112

7.4.1.1 I/O Channels

Bit Assignment - PAE Input Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Terminal	X0 07	X0 06	X0 05	X0 04	X0 03	X0 02	X0 01	X0 00
Byte	1							
Bit	7	6	5	4	3	2	1	0
Terminal	X1 07	X1 06	X1 05	X1 04	X1 03	X1 02	X1 01	X1 00
Byte	2							
Bit	7	6	5	4	3	2	1	0
Terminal	X2 07	X2 06	X2 05	X2 04	X2 03	X2 02	X2 01	X2 00
Byte	3							
Bit	7	6	5	4	3	2	1	0
Terminal	X3 07	X3 06	X3 05	X3 04	X3 03	X3 02	X3 01	X3 00

Tab. 219: PAE input data

7.4.1.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.4.2 Cube20 DO16 Art. No. 56117

7.4.2.1 I/O Channels

Bit Assignment - PAA Output Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Terminal	X1 03	X1 02	X1 01	X1 00	X0 03	X0 02	X0 01	X0 00
Byte	1							
Bit	7	6	5	4	3	2	1	0
Terminal	X3 03	X3 02	X3 01	X3 00	X2 03	X2 02	X2 01	X2 00

Tab. 220: PAA output data

7.4.2.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.4.3 Cube20 DO32 Art. No. 56118

7.4.3.1 I/O Channels

Bit Assignment - PAA Output Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Terminal	X0 07	X0 06	X0 05	X0 04	X0 03	X0 02	X0 01	X0 00
Byte	1							
Bit	7	6	5	4	3	2	1	0
Terminal	X1 07	X1 06	X1 05	X1 04	X1 03	X1 02	X1 01	X1 00
Byte	2							
Bit	7	6	5	4	3	2	1	0
Terminal	X2 07	X2 06	X2 05	X2 04	X2 03	X2 02	X2 01	X2 00
Byte	3							
Bit	7	6	5	4	3	2	1	0
Terminal	X3 07	X3 06	X3 05	X3 04	X3 03	X3 02	X3 01	X3 00

Tab. 221: PAA output data

7.4.3.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.4.4 Cube20 DI32 NPN / PNP Art. No. 56121

7.4.4.1 I/O Data

Bit Assignment - PAE Input Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Terminal	X0 07	X0 06	X0 05	X0 04	X0 03	X0 02	X0 01	X0 00
Byte	1							
Bit	7	6	5	4	3	2	1	0
Terminal	X1 07	X1 06	X1 05	X1 04	X1 03	X1 02	X1 01	X1 00
Byte	2							
Bit	7	6	5	4	3	2	1	0
Terminal	X2 07	X2 06	X2 05	X2 04	X2 03	X2 02	X2 01	X2 00
Byte	3							
Bit	7	6	5	4	3	2	1	0
Terminal	X3 07	X3 06	X3 05	X3 04	X3 03	X3 02	X3 01	X3 00

Tab. 222: PAE input data

7.4.4.2 Parameterization



NOTE

The parameter bytes of this module are reserved.

7.4.5 Cube20 DI16 DO16 Art. No. 56168

7.4.5.1 I/O Data

Bit Assignment - PAE Input Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Terminal	X2 07	X2 06	X2 05	X2 04	X2 03	X2 02	X2 01	X2 00
Byte	1							
Bit	7	6	5	4	3	2	1	0
Terminal	X3 07	X3 06	X3 05	X3 04	X3 03	X3 02	X3 01	X3 00

Tab. 223: PAE input data

Bit Assignment - PAA Output Data

Byte	0							
Bit	7	6	5	4	3	2	1	0
Terminal	X0 07	X0 06	X0 05	X0 04	X0 03	X0 02	X0 01	X0 00
Byte	1							
Bit	7	6	5	4	3	2	1	0
Terminal	X1 07	X1 06	X1 05	X1 04	X1 03	X1 02	X1 01	X1 00

Tab. 224: PAA output data

7.4.5.2 Parameterization

**NOTE**

The parameter bytes of this module are reserved.

7.4.6 Cube20 AI4 U/I Art. No. 56200

7.4.6.1 Parameterization

Parameter Bytes

**NOTE**

Parameter length: 12 Byte

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	
3	Settings Channel 1
4	Reserved
5	
6	Settings Channel 2
7	Reserved
8	
9	Settings Channel 3
10	Reserved
11	

13

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 00: Measuring range	000	disabled
		001	0 to 10 V
		010	±10 V
		011	4 to 20 mA
		100	0 to 20 mA
0x40yy:02	Channel 00: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 00: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 00: Smoothing	00	No
		01	weak
		10	medium
		11	strong
0x40yy:05	Channel 01: Measuring range	Same as Subindex (SI) 01	
0x40yy:06	Channel 01: Diagnosis	Same as SI 02	
0x40yy:07	Channel 01: Data format	Same as SI 03	
0x40yy:08	Channel 01: Smoothing	Same as SI 04	
0x40yy:09	Channel 02: Measuring range	Same as SI 01	
0x40yy:0A	Channel 02: Diagnosis	Same as SI 02	
0x40yy:0B	Channel 02: Data format	Same as SI 03	
0x40yy:0C	Channel 02: Smoothing	Same as SI 04	
0x40yy:0D	Channel 03: Measuring range	Same as SI 01	
0x40yy:0E	Channel 03: Diagnosis	Same as SI 02	
0x40yy:0F	Channel 03: Data format	Same as SI 03	
0x40yy:10	Channel 03: Smoothing	Same as SI 04	

Tab. 225: *Parameter overview*

7.4.6.2 Explanation of Smoothing

Device prefilter

The analog signal can be prefiltered in the device to reduce sensitivity to transient fluctuations in measured values, for example.

Measuring time per channel: 4,0 ms.

Total cycle time: Measuring time per channel + 4.0 ms.

Chart for time-related smoothing behavior

The chart below shows module behavior to a step response in an active channel.

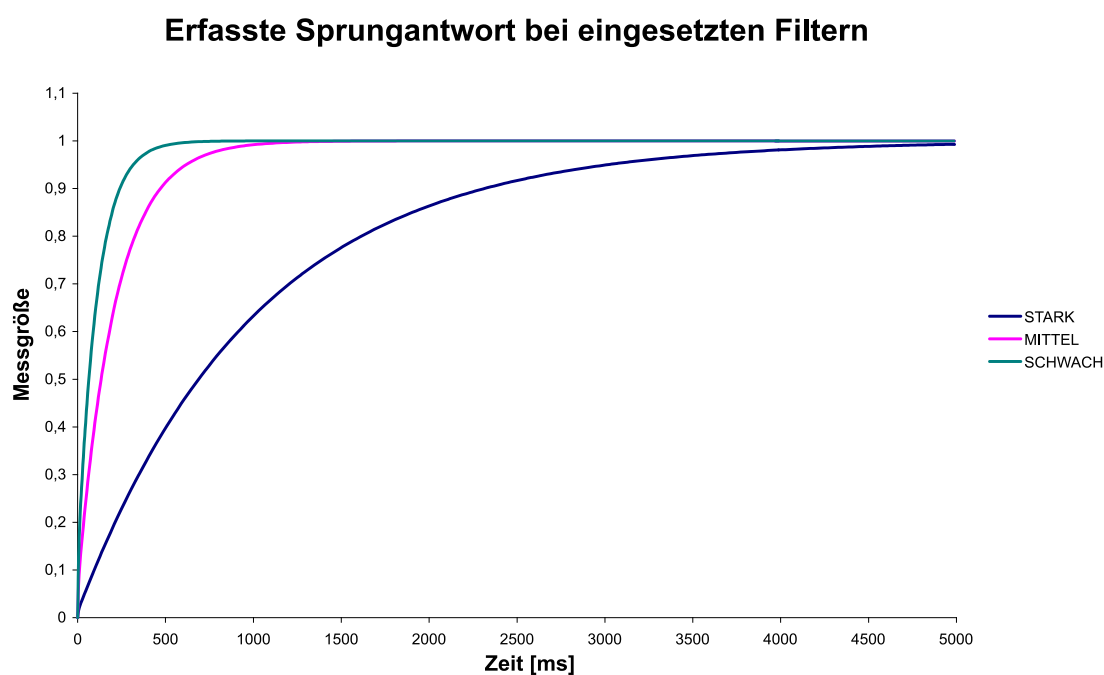


Fig. 34 Smoothing time behavior

7.4.6.3 Binary Display of Analog Data

Bit Assignment of I/O Data in Intel Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8
Byte	LOWBYTE								VZ	HIGHBYTE						
	7									6						

Tab. 226: Bit assignment of analog I/O data in Intel format

Bit Assignment of I/O Data in Motorola Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

<

Tab. 227: Bit assignment of analog I/O data in Motorola format

7.4.6.4 Display of Analog Values

Display of Analog Values of 16-bit Input Data with VZ

Measuring range	Measured value			Range	Diagnosis
	Decimal	Hexadecimal	Binary		
0 to 10 V					
> 11.7589 V	32767	7FFF	0111 1111 1111 1111	Overload	Yes
11.7589 V	32511	7EFF	0111 1110 1111 1111		Yes
⋮	⋮	⋮	⋮		As of limit
10.0004 V	27649	6C01	0110 1100 0000 0001		No
10 V	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
5 V	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
0.0004 V	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		
< 0	0	0000	0000 0000 0000 0000	Underload	As of limit

Tab. 228: Analog values within range 0 to 10 V

Measuring range	Measured value			Range	Diagnosis
-10 to +10 V	Decimal	Hexadecimal	Binary		
> 11.7589 V	32767	7FFF	0111 1111 1111 1111	Overload	Yes
11.7589 V	32511	7EFF	0111 1110 1111 1111		Yes
⋮	⋮	⋮	⋮		As of limit
10.0004 V	27649	6C01	0110 1100 0000 0001		No
10 V	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
5 V	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
0.0004 V	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		
-0.0004 V	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-5 V	-13824	CA00	1100 1010 0000 0000		
⋮	⋮	⋮	⋮		
-10 V	-27648	9400	1001 0100 0000 0000		
-10.0004 V	-27649	93FF	1001 0011 1111 1111	Underload	No
⋮	⋮	⋮	⋮		As of limit
-11.759 V	-32512	8100	1000 0001 0000 0000		Yes
< -11.759 V	-32768	8000	1000 0000 0000 0000		Yes

Tab. 229: Analog values within range -10 to +10 V

Measuring range	Measured value			Range	Diagnosis
4 to 20 V	Decimal	Hexadecimal	Binary		
>22.8142 mA	32767	7FFF	0111 1111 1111 1111	Overload	Yes
22.8142 mA	32511	7EFF	0111 1100 0000 0000		Yes
⋮	⋮	⋮	⋮		As of limit
20.0006 mA	27649	6C01	0110 1100 0000 0001		No
20 mA	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
12 mA	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
4.0006 mA	1	0001	0000 0000 0000 0001		
4 mA	0	0000	0000 0000 0000 0000		
<4 mA	0	0000	0000 0000 0000 0000	Underload	As of limit

Tab. 230: Analog values within range 4 to 20 mA

Measuring range	Measured value			Range	Diagnosis
0 to 20 V	Decimal	Hexadecimal	Decimal		
>22,963 mA	32767	7FFF	0111 1111 1111 1111	Overload	Yes
22.963 mA	31744	7C00	0111 1110 1111 1111		Yes
⋮	⋮	⋮	⋮		As of limit
20.0007 mA	27649	6C01	0110 1100 0000 0001		No
20 mA	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
10 mA	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
0.0007 mA	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		
< 0	0	0000	0000 0000 0000 0000	Underload	No

Tab. 231: Analog values within range 0 to 20 mA

7.4.7 Cube20 AO4 U/I Art. No. 56220

7.4.7.1 Parameterization

Parameter Bytes


NOTE

Parameter length: 16 bytes

Byte	Meaning
0	Settings Channel 0
1	Reserved
2	
3	
4	Settings Channel 1
5	Reserved
6	
7	
8	Settings Channel 2
9	Reserved
10	
11	
12	Settings Channel 3
13	Reserved
14	
15	

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Channel 00: Measuring range	000	disabled
		001	0 to 10 V
		010	±10 V
		011	4 to 20 mA
		100	0 to 20 mA
0x40yy:02	Channel 00: Diagnosis	0	Signal
		1	Do not signal
0x40yy:03	Channel 00: Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:04	Channel 01: Measuring range	Same as Subindex (SI) 01	
0x40yy:05	Channel 01: Diagnosis	Same as SI 02	
0x40yy:06	Channel 01: Data format	Same as SI 03	
0x40yy:07	Channel 02: Measuring range	Same as SI 01	
0x40yy:08	Channel 02: Diagnosis	Same as SI 02	
0x40yy:09	Channel 02: Data format	Same as SI 03	

Index	Parameters	Value	Description
0x40yy:0A	Channel 03: Measuring range	Same as SI 01	
0x40yy:0B	Channel 03: Diagnosis	Same as SI 02	
0x40yy:0C	Channel 03: Data format	Same as SI 03	

Tab. 232: *Parameter overview*

7.4.7.2 Binary Display of Analog Data

Bit Assignment of I/O Data in Intel Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8
Byte	LOWBYTE							VZ	HIGHBYTE							
	7								6							

Tab. 233: *Bit assignment of analog I/O data in Intel format*

Bit Assignment of I/O Data in Motorola Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte	VZ	HIGHBYTE							LOWBYTE							
	7								6							

Tab. 234: *Bit assignment of analog I/O data in Motorola format*

7.4.7.3 Display of Analog Values

Display of Analog Values of 16-bit Input Data with VZ

Measuring range	Measured value			Range	Diagnosis
0 to 10 V	Decimal	Hexadecimal	Binary		
10 V	≥27649	≥6C01	≥0110 1100 0000 0001	Overload	No
10 V	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
5 V	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
0,0004 V	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		
0	< 0	< 0000	< 0000 0000 0000 0000	Underload	No

Tab. 235: Analog values within range 0 to 10 V

Measuring range	Measured value			Range	Diagnosis
-10 to +10 V	Decimal	Hexadecimal	Binary		
10 V	≥27649	≥6C01	≥0110 1100 0000 0001	Overload	No
10 V	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
5 V	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
0,0004 V	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		
-0,0004 V	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-5 V	-13824	CA00	1100 1010 0000 0000		
⋮	⋮	⋮	⋮		
-10 V	-27648	9400	1001 0100 0000 0000		
-10 V	≤-27649	≤93FF	≤1001 0011 1111 1111	Underload	No

Tab. 236: Analog values within range -10 to +10 V

Measuring range	Measured value			Range	Diagnosis
4 to 20 V	Decimal	Hexadecimal	Binary		
20 mA	≥27649	≥6C01	≥0110 1100 0000 0001	Overload	No
20 mA	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
12 mA	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
4.0006 mA	1	0001	0000 0000 0000 0001		
4 mA	0	0000	0000 0000 0000 0000		

Tab. 237: Analog values within range 4 to 20 mA

Measuring range	Measured value			Range	Diagnosis
0 to 20 V	Decimal	Hexadecimal	Decimal		
20 mA	≥27649	≥6C01	≥0110 1100 0000 0001	Overload	No
20 mA	27648	6C00	0110 1100 0000 0000	Nominal range	No
⋮	⋮	⋮	⋮		
10 mA	13824	3600	0011 0110 0000 0000		
⋮	⋮	⋮	⋮		
0.0007 mA	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		

Tab. 238: Analog values within range 0 to 20 mA

7.4.8 Cube20 AI4 RTD Art. No. 56230

7.4.8.1 Parameterization

Parameter Bytes


NOTE

Parameter length: 5 Byte

Byte	Meaning
0	Settings Channel 0 Data format, temperature unit, channel diagnosis
1	Measuring Channel 0 and sensor Channel 0
2	Measuring Channel 1 and sensor Channel 1
3	Measuring Channel 2 and sensor Channel 2
4	Measuring Channel 3 and sensor Channel 3

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:02	Temperature unit	0	°C (Celsius)
		1	°F (Fahrenheit)
0x40yy:03	Channel diagnosis	0	Signal
		1	Do not signal
0x40yy:04	Channel 00: Measuring channel	00	disabled
		01	3-wire connection
		10	2-wire connection
0x40yy:05	Channel 00: Sensor	0000	Platinum 100
		0001	Platinum 200
		0010	Platinum 500
		0011	Platinum 1000
		0101	Nickel 120
		0110	Nickel 200
		0111	Nickel 500
		1000	Nickel 1000
		1001	Resistance
0x40yy:06	Channel 01: Measuring channel	Same as SI 04	
0x40yy:07	Channel 01: Sensor	Same as SI 05	
0x40yy:08	Channel 02: Measuring channel	Same as SI 04	
0x40yy:09	Channel 02: Sensor	Same as SI 05	
0x40yy:0A	Channel 03: Measuring channel	Same as SI 04	
0x40yy:0B	Channel 03: Sensor	Same as SI 05	

Tab. 239: Parameter overview

7.4.8.2 Binary Display of Analog Data

Bit Assignment of I/O Data in Intel Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8
Byte	LOWBYTE								VZ	HIGHBYTE						
	7									6						

Tab. 240: Bit assignment of analog I/O data in Intel format

Bit Assignment of I/O Data in Motorola Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte	VZ	HIGHBYTE							LOWBYTE							
		7								6						

Tab. 241: Bit assignment of analog I/O data in Motorola format

7.4.8.3 Display of Analog Values

Display of analog values of 16-bit input data with VZ

Analog values are displayed as follows:

Range	Steps
Resistance range	1/10 Ω
Temperature range	1/10°C (Celsius) or 1/10°F (Fahrenheit)

Analog Values for Platinum, Unit: °C (Celsius)

Measuring range	Measured value			Range	Diagnosis
Pt100 / Pt200 Pt500 / Pt1000	Decimal	Hexadecimal	Binary		
>850°C	8500	2134	0010 0001 0011 0100	Overload	As of limit
850°C	8500	2134	0010 0001 0011 0100	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0001	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-200°C	-2000	F830	1111 1000 0011 0000	Underload	As of limit
<-200°C	-2000	F830	1111 1000 0011 0000		

Tab. 242: Analog values within platinum range / °C

Analog Values for Nickel, Unit: °C (Celsius)

Measuring range	Measured value			Range	Diagnosis
Ni100 / Ni200 Ni500 / Ni1000	Decimal	Hexadecimal	Binary		
>250°C	2500	09C4	0000 1001 1100 0100	Overload	As of limit
250°C	2500	09C4	0000 1001 1100 0100	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0000	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-60°C	-600	FDA8	1111 1101 1010 1000	Underload	As of limit
<-60°C	-600	FDA8	1111 1101 1010 1000		

Tab. 243: Analog values within nickel range / °C

Analog Values for Platinum: Unit: °F (Fahrenheit)

Measuring range Pt100 / Pt200 Pt500 / Pt1000	Measured value			Range	Diagnosis
	Decimal	Hexadecimal	Binary		
>1562°F	15620	3D04	0011 1101 0000 0100	Overload	As of limit
1562°F	15620	3D04	0011 1101 0000 0100	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	1	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-328°F	-3280	F330	1111 0011 0011 0000	Underload	As of limit
<-328°F	-3280	F330	1111 0011 0011 0000		

Tab. 244: Analog values within platinum range / °F

Analog Values for Nickel: Unit: °F (Fahrenheit)

Measuring range Ni100 / Ni200 Ni500 / Ni1000	Measured value			Range	Diagnosis
	Decimal	Hexadecimal	Binary		
>482°F	4820	12D4	0001 0010 1101 0100	Overload	As of limit
482°F	4820	12D4	0001 0010 1101 0100	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	1	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-76°F	-760	FD08	1111 1101 0000 1000	Underload	As of limit
<-76°F	-760	FD08	1111 1101 0000 1000		

Tab. 245: Analog values within nickel range / °F

Analog Values for Resistance Measurement [Ω]

Measuring range	Measured value			Range	Diagnosis
Resistance	Decimal	Hexadecimal	Binary		
>3000 Ω	30000	7530	0111 0101 0011 0000	Overload	As of limit
3000 Ω	30000	7530	0111 0101 0011 0000	Nominal range	No
:	:	:	:		
0,1 Ω	1	0001	0000 0000 0000 0001		
0	0	0000	0000 0000 0000 0000		

Tab. 246: Analog values within resistance range

7.4.9 Cube20 AI4 TH Art. No. 56240**7.4.9.1 Parameterization****Parameter Bytes****NOTE**

Parameter length: 5 Byte

Byte	Meaning
0	Settings Channel 0 Data format, temperature unit, channel diagnosis
1	Measuring channel 0
2	Measuring channel 1
3	Measuring channel 2
4	Measuring channel 3

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	Data format	0	Byte Sequence High/Low (Motorola)
		1	Byte Sequence Low/High (Intel)
0x40yy:02	Temperature unit	0	°C (Celsius)
		1	°F (Fahrenheit)
0x40yy:03	Channel diagnosis	0	Signal
		1	Do not signal
0x40yy:04	Measuring channel 0	000	disabled
		001	Type K
		010	Type N
		011	Type J
		100	Type E
		101	Type R
0x40yy:05	Measuring channel 1	Same as Subindex (SI) 04	
0x40yy:06	Measuring channel 2	Same as SI 04	
0x40yy:07	Measuring channel 3	Same as SI 04	

Tab. 247: *Parameter overview*

7.4.9.2 Binary Display of Analog Data

Bit Assignment of I/O Data in Intel Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8
Byte	LOWBYTE								VZ	HIGHBYTE						
	7									6						

Tab. 248: *Bit assignment of analog I/O data in Intel format*

Bit Assignment of I/O Data in Motorola Format

I/O Data															
Byte	7	6	5	4	3	2	1	0							
Channel	3			2			1			0					

Bit	MSB							LSB	MSB							LSB
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte	VZ	HIGHBYTE							LOWBYTE							
		7								6						

Tab. 249: *Bit assignment of analog I/O data in Motorola format*

7.4.9.3 Display of Analog Values

Display of analog values of 16-bit input data with VZ

Analog values are displayed as follows:

Range	Steps
Temperature range	1/10°C (Celsius) or 1/10°F (Fahrenheit)

Unit °C (Celsius)

Analog Values for Type K

Measuring range Type K	Measured value			Range	Diagnosis
	Decimal	Hexadecimal	Binary		
>1372°C	13720	3598	0011 0101 1001 1000	Overload	As of limit
1372°C	13720	3598	0011 0101 1001 1000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0001	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-200°C	-2000	F830	1111 1000 0011 0000	Underload	As of limit
<-200°C	-2000	F830	1111 1000 0011 0000		

Tab. 250: Analog values for Type K / °C

Analog Values for Type N

Measuring range Type N	Measured value			Range	Diagnosis
	Decimal	Hexadecimal	Binary		
>1300°C	13000	32C8	0011 0010 1100 1000	Overload	As of limit
1300°C	13000	32C8	0011 0010 1100 1000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0001	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-200°C	-2000	F830	1111 1000 0011 0000	Underload	As of limit
<-200°C	-2000	F830	1111 1000 0011 0000		

Tab. 251: Analog values for Type N / °C

Analog Values for Type J

Measuring range	Measured value			Range	Diagnosis
Type J	Decimal	Hexadecimal	Binary		
>1200°C	12000	2EE0	0010 1110 1110 0000	Overload	As of limit
1200°C	12000	2EE0	0010 1110 1110 0000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0001	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-200°C	-2000	F830	1111 1000 0011 0000	Underload	As of limit
<-200°C	-2000	F830	1111 1000 0011 0000		

Tab. 252: Analog values for Type J / °C

Analog Values for Type E

Measuring range	Measured value			Range	Diagnosis
Type E	Decimal	Hexadecimal	Binary		
>1000°C	10000	2710	0010 0111 0001 0000	Overload	As of limit
1000°C	10000	2710	0010 0111 0001 0000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0001	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-200°C	-2000	F830	1111 1000 0011 0000	Underload	As of limit
<-200°C	-2000	F830	1111 1000 0011 0000		

Tab. 253: Analog values for Type E / °C

Analog Values for Type R

Measuring range	Measured value			Range	Diagnosis
Type R	Decimal	Hexadecimal	Binary		
>1768°C	17680	4510	0100 0101 0001 0000	Overload	As of limit
1768°C	17680	4510	0100 0101 0001 0000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°C	0	0001	0000 0000 0000 0001		
0°C	0	0000	0000 0000 0000 0000		
-0.1°C	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-50°C	-500	FE0C	1111 1110 0000 1100	Underload	As of limit
<-50°C	-500	FE0C	1111 1110 0000 1100		

Tab. 254: Analog values for Type R / °C

Unit °F (Fahrenheit)**Analog Values for Type K**

Measuring range	Measured value			Range	Diagnosis
Type K	Decimal	Hexadecimal	Binary		
>2501.6°F	25016	61B8	0011 0000 11011 1000	Overload	As of limit
2501.6°F	25016	61B8	0011 0000 11011 1000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	0	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-328°F	-3280	F330	1111 0011 0011 0000	Underload	As of limit
<-328°F	-3280	F330	1111 0011 0011 0000		

Tab. 255: Analog values for Type K / °F

Analog Values for Type N

Measuring range	Measured value			Range	Diagnosis
Type N	Decimal	Hexadecimal	Binary		
>2372°F	23720	5CA8	0101 1100 1010 1000	Overload	As of limit
2372°F	23720	5CA8	0101 1100 1010 1000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	0	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-328°F	-3280	F330	1111 0011 0011 0000	Underload	As of limit
<-328°F	-3280	F330	1111 0011 0011 0000		

Tab. 256: Analog values for Type N / °F

Analog Values for Type J

Measuring range	Measured value			Range	Diagnosis
Type J	Decimal	Hexadecimal	Binary		
>2192°F	21920	55A0	0101 0101 1010 0000	Overload	As of limit
2192°F	21920	55A0	0101 0101 1010 0000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	0	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-328°F	-3280	F330	1111 0011 0011 0000	Underload	As of limit
<-328°F	-3280	F330	1111 0011 0011 0000		

Tab. 257: Analog values for Type J / °F

Analog Values for Type E

Measuring range	Measured value			Range	Diagnosis
Type E	Decimal	Hexadecimal	Binary		
>1832°F	18320	4790	0100 0111 1001 0000	Overload	As of limit
1832°F	18320	4790	0100 0111 1001 0000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	0	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-328°F	-3280	F330	1111 0011 0011 0000	Underload	As of limit
<-328°F	-3280	F330	1111 0011 0011 0000		

Tab. 258: Analog values for Type E / °F

Analog Values for Type R

Measuring range	Measured value			Range	Diagnosis
Type R	Decimal	Hexadecimal	Binary		
>3214.4°F	32144	7D90	0111 1101 1001 0000	Overload	As of limit
3214.4°F	32144	7D90	0111 1101 1001 0000	Nominal range	No
⋮	⋮	⋮	⋮		
0.1°F	0	0001	0000 0000 0000 0001		
0°F	0	0000	0000 0000 0000 0000		
-0.1°F	-1	FFFF	1111 1111 1111 1111		
⋮	⋮	⋮	⋮		
-58°F	-580	FDBC	1111 1101 1011 1100	Underload	As of limit
<-58°F	-580	FDBC	1111 1101 1011 1100		

Tab. 259: Analog values for Type R / °F

7.4.10 Cube20 BN67 DIO8 – Art. No. 56450

7.4.10.1 I/O Data

Bit Assignment - PAE Input Data

Byte	0 (Terminal X3)				0 (Terminal X2)			
Bit	7	6	5	4	3	2	1	0
Channel	03	02	01	00	03	02	01	00

Tab. 260: PAE input data

Bit Assignment - PAA Output Data

Byte	0 (Terminal X3)				0 (Terminal X2)			
Bit	7	6	5	4	3	2	1	0
Channel	03	02	01	00	03	02	01	00

Tab. 261: PAA input data

7.4.10.2 Parameterization

Parameter Bytes



NOTE

Parameter length: 1 byte

Parameter Overview

Index	Parameters	Value	Description
0x40yy:01	X2:00 (Ch00)	0	Input with NO function
		1	Output
0x40yy:02	X2:01 (Ch01)	Same as Subindex (SI) 01	
0x40yy:03	X2:02 (Ch02)	Same as SI 01	
0x40yy:04	X2:03 (Ch03)	Same as SI 01	
0x40yy:05	X3:00 (Ch04)	Same as SI 01	
0x40yy:06	X3:01 (Ch05)	Same as SI 01	
0x40yy:07	X3:02 (Ch06)	Same as SI 01	
0x40yy:08	X3:03 (Ch07)	Same as SI 01	

Tab. 262: Parameter overview

7.5 Accessories



A list of Cube67+ accessories is contained in the Cube67+ System Manual Art. No. 56974.

Information on accessories is available in our catalog and our online shop at:

www.onlineshop.murrelektronik.com

7.5.1 Grounding Strap

Grounding strap M4



Art. No. 4000-71001-0410004

For Cube67 bus nodes

Cross-section 4 mm²

100 mm for M4 hole

Screws

for grounding strap M4



Art. No. 4000-71003-0101604

7.5.2 System Components

Screw plugs 7/8"

Screw plug, metal

Art. No. 55390

Screw plug, plastic

Art. No. 55385

Screw plug M12

Screw plug, shielded

Art. No. 996049

Screw plug, unshielded

Art. No. 58627

Terminating resistors

Terminating resistor M12,
A-coded, 6-pin

Art. No. 7000-15041-0000000

7.5.3 Tools

Torque wrench

M12 plug

Art. No. 7000-99102-000 0000

7/8" plug

Art. No. 7000-99104-000 0000

8 Annex 2: Additional Information

8.1 Glossary

BN-EC	Bus Node EtherCAT
Byte	Equivalent to 8 bits
CH	Channel
DI	Digital Input
DIO	Digital Input/Output
DIN	Deutsches Institut für Normung (German Standards Institute)
DO	Digital Output
I/O	Input/Output
EC Directive 2004/108/EC	EMC Directive.
EMC	Electromagnetic Compatibility.
EN	European Standard
ESD	Electrostatic Discharge
ESI Files	EtherCAT Slave Information
ESM	EtherCAT State Machine The state of EtherCAT slaves is controlled by the EtherCAT State Machine. Depending on the state, different functions are accessible or executable in the EtherCAT slave. Specific commands must be sent from the EtherCAT Master to the device, in particular when the slave runs up.
FE	Function ground
EtherCAT master	The EtherCAT master is the I/O controller. It must support MDP (Modular Device Profile).
I	Current
IEC 61188	IEC 61188 Type 12 (EtherCAT) open protocol
IGMP	Internet Group Management Protocol
IP67	Ingress Protection, protection degree to DIN EN 60529 1st digit = protection against contact and foreign bodies 2nd digit = protection against water 6: Dust-proof, protection against access with a wire 7: Protection against temporary immersion
ISO	International Standard Organization
LED	Light Emitting Diode
MDP	Modular Device Profile
PELV	Protective Extra Low Voltage

Segment	Left segment of the internal system connection (Sockets 0 and 2) and right segment of the system link (Sockets 1 and 3)
SELV	Safety Extra Low Voltage.
U	voltage
U/I	Voltage / current

8.2 Legal Provisions

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